

# Supplementary Information

**An upstream placebo shows that reservoir siting explains the apparent drought buffering of seasonal reservoirs in Chile’s megadrought**

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This Supplementary Information collects the supporting design, robustness, and diagnostic results referenced in the main text, numbered in the order the main text first cites them.

Supplementary Figure S1 is the water-rights induced-demand test and Supplementary Figure S2 the non-parametric aridity common-support check; Supplementary Figures S3 and S4 give the full study-area map and the matching diagnostics; and Supplementary Figure S5 documents the drought forcing, Supplementary Figure S6 the siting-confound decomposition, Supplementary Figure S7 the barren-land evapotranspiration (ET) confound demonstration, and Supplementary Figure S8 the informativeness of the null.

Supplementary Table S1 tests reservoir-use heterogeneity (irrigation-only subset), Supplementary Table S2 collects the inter-basin spillover (cuenca disjointness) and administrative-demand checks including the treated-versus-control closure test, and Supplementary Table S3 screens the matched controls against the 1,370-dam national inventory and re-fits the streamflow results with a re-balanced dam-free control pool. The storage-band series follows: Supplementary Table S4 re-fits the trends weighted by reservoir capacity and in absolute fleet volumes, Supplementary Table S5 on the coverage-stable reservoir subset, and Supplementary Table S6 confirms every trend, step, and drift coefficient with a null-imposed wild cluster bootstrap, the small-cluster refinement appropriate to the 22 to 26 reservoir clusters. Supplementary Tables S7 to S9 harden the carryover contrast with the ratio-denominator sensitivity (unregulated upstream flow), the strictly-paired contrast, and the hydropower-excluded and leave-one-out inference.

Supplementary Table S10 quantifies the drought-conditional structure of streamflow missingness, Supplementary Table S11 re-derives the streamflow results with the Standardized Streamflow Index (SSI) standardized on the 1990–2020 baseline, and Supplementary Table S12 gives the volumetric-scale (log-flow) checks. Supplementary Tables S13 and S14 report the winsorization-threshold sensitivity and the rank-based tests on raw, uncapped water-rights volumes. Supplementary Table S15 tests the snowmelt confound on the differential estimand of the upstream/downstream placebo with standardized ERA5-Land snow water equivalent (snow-adjusted refits, a low-snow subsample, snow-year modulation, and an early/late stability split), Supplementary Table S16 collects the relegated ET outcomes, and Supplementary Table S17 bounds the groundwater-substitution confound with the Dirección General de Aguas (DGA) well hydrograph network.

The matching diagnostics run from Supplementary Table S18 to Supplementary Table S26: the pre-drought aridity-window sensitivity, the aridity-overlap sensitivity and its fully non-parametric

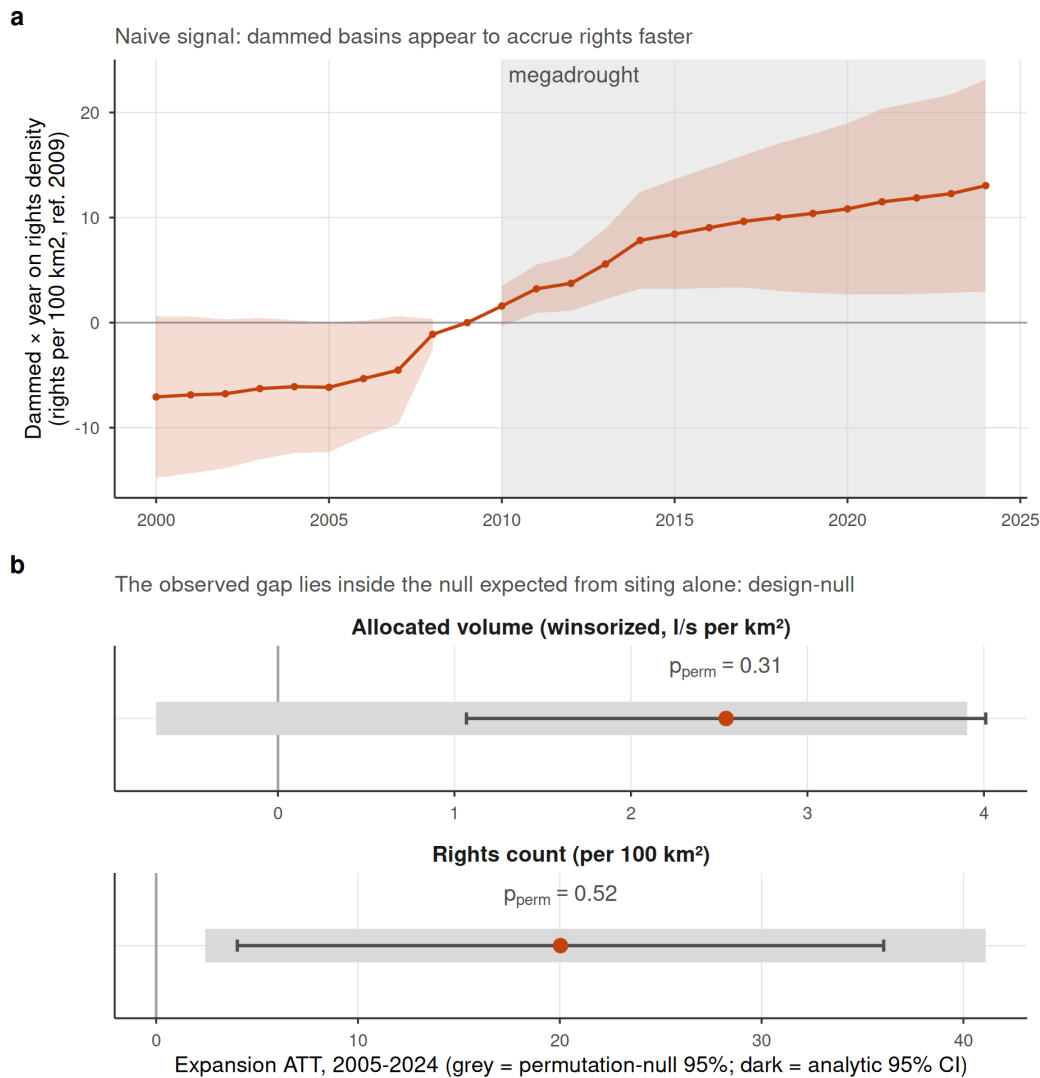
re-match, the metric and functional-form checks, the non-linear aridity-by-forcing sensitivity, the strict hard-balanced fit (effective control sample 19) with its weight diagnostics and leave-one-out sensitivities, the smooth cubic national time trend, and the physical-volume translation of the equivalence margin. Supplementary Table S27 assesses spatial autocorrelation and reports spatially-restricted inference, and Supplementary Table S28 the permutation inference for the within-region (siting-ladder rung 4) residual.

The placebo battery spans Supplementary Tables S29 to S46: the direct within-basin contrast and its elevation-partitioned permutation null; the comparability audit with its catchment-area, elevation-adjusted, cumulative-scale (discharge-adjusted), and geomorphological checks; the gauge-classification robustness, the routed-network (HydroRIVERS) validation, and the cascade-dam contamination screen against the 1,370-dam inventory; the non-linear (quadratic) and dry-tail transmission tests; the early/late megadrought phase split and its permutation inference; the downstream extraction-masking bound, the placebo net of an interaction between the Standardized Precipitation Evapotranspiration Index (SPEI) and local cropland, and the sub-watershed SPEI exposure-misclassification bound; and the paired contrast's own power.

Supplementary Tables S47 to S49 give the siting-confound decomposition ladder in numeric form, the positive control on the streamflow estimator, and the equivalence and minimum detectable effect bounds on the headline nulls. Supplementary Table S50 reports the storage-capacity (carryover) heterogeneity of the buffering test, Supplementary Table S51 the minimum detectable effect for the cropland pre-trend, Supplementary Tables S52 and S53 the storage-band trends with the megadrought level shift and its step-versus-trend decomposition, and Supplementary Table S54 the carryover capacity of the treated reservoirs. All items are regenerated by the **targets** pipeline.

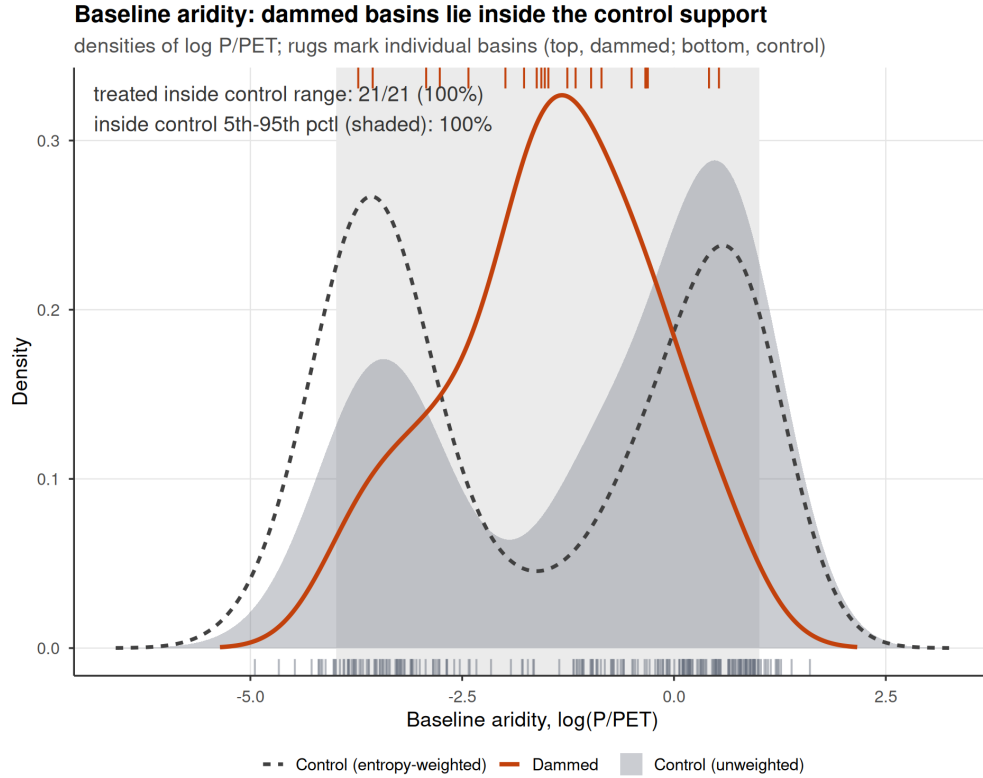
## Supplementary Figures

### Water rights: no induced demand survives once siting is matched out



**Supplementary Figure S1 | Water rights show no induced demand once siting is matched out.** (a) Event study of cumulative consumptive water-rights density (dammed x year, ref. 2009; rights per 100 km²): the dammed-versus-control gap appears to widen through the megadrought, the naive induced-demand signal. (b) For both the rights count and the winsorized allocated volume, the expansion average treatment effect on the treated (ATT; 2005–2024; orange point, dark analytic 95% CI) is positive and its analytic interval excludes zero, yet it lies squarely inside the 95% band of the permutation null (grey), what random treatment assignment within climate x aridity strata already produces given the dammed basins' markedly higher aridity, so it is null under randomization inference ( $p_{\text{perm}} = 0.52$  for counts and 0.31 for the winsorized volume, both estimated on the full matched sample with the 99.5th-percentile cap under unit-level stratified permutation; the threshold sensitivity is Supplementary Table S13, and the aridity-overlap subset with its spatial-block inference, a different sample and permutation scheme, is Supplementary

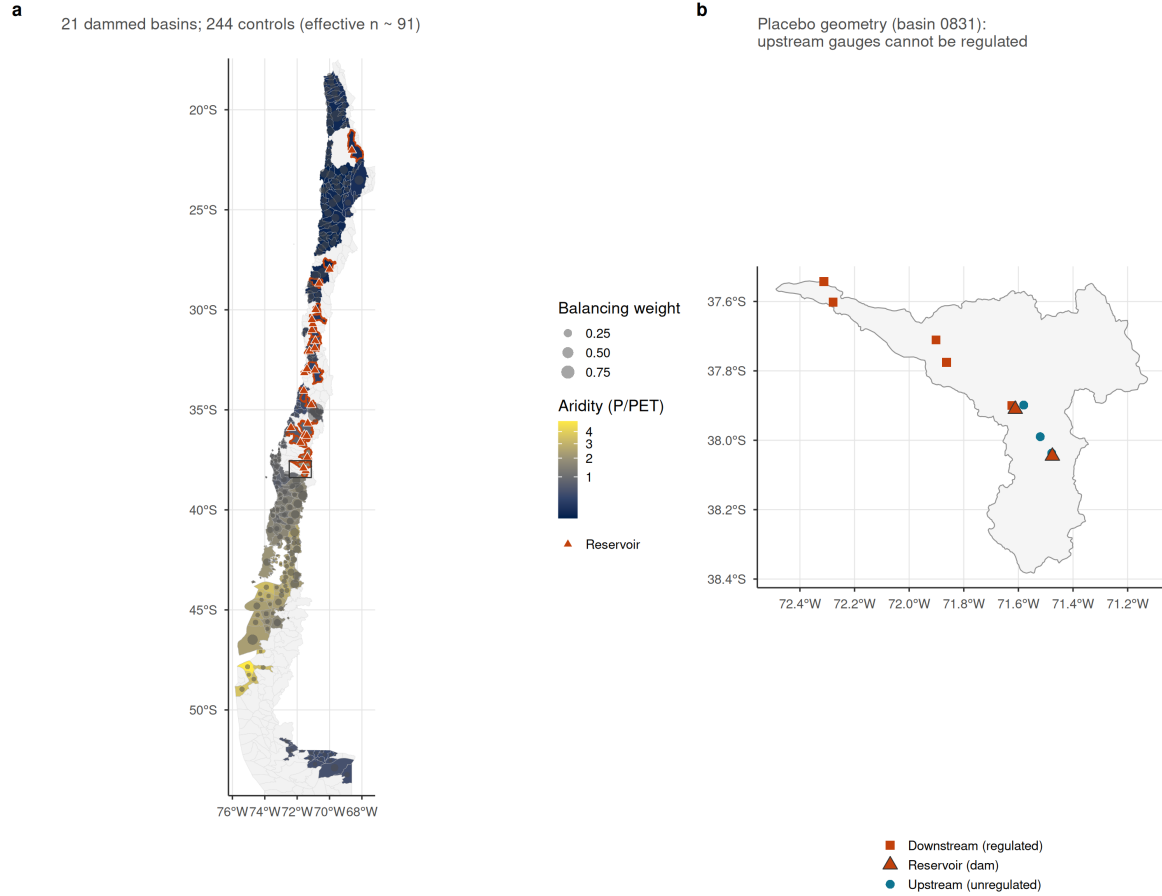
Table S27). A conventional matched analysis of this registry would report induced demand; the design-based inference reveals it as siting.



**Supplementary Figure S2 | Baseline aridity: non-parametric common support.** Densities of long-term baseline aridity,  $\log(P/PET)$ , for control basins (grey area, unweighted; dashed line, entropy-weighted) and dammed basins (orange), with per-basin rug marks (top, dammed; bottom, control). The dammed basins are systematically more arid than the unweighted control mean, the residual imbalance (SMD 0.17) that the doubly-robust outcome model adjusts parametrically; this plot shows the adjustment is interpolation, not extrapolation. All 21 dammed basins lie inside the control range and inside the control 5th–95th percentile band (shaded), and the control density is strictly positive across the entire treated support, so no treated unit relies on out-of-support extrapolation. The hard-balanced sensitivity (Supplementary Table S23), which removes the residual imbalance non-parametrically at the cost of an effective control sample of 19, reaches the same null, so the conclusion does not rest on the parametric adjustment.

### A national matched design across Chile's aridity gradient

Dammed basins sit in the arid centre; weighted controls span the same climate strata



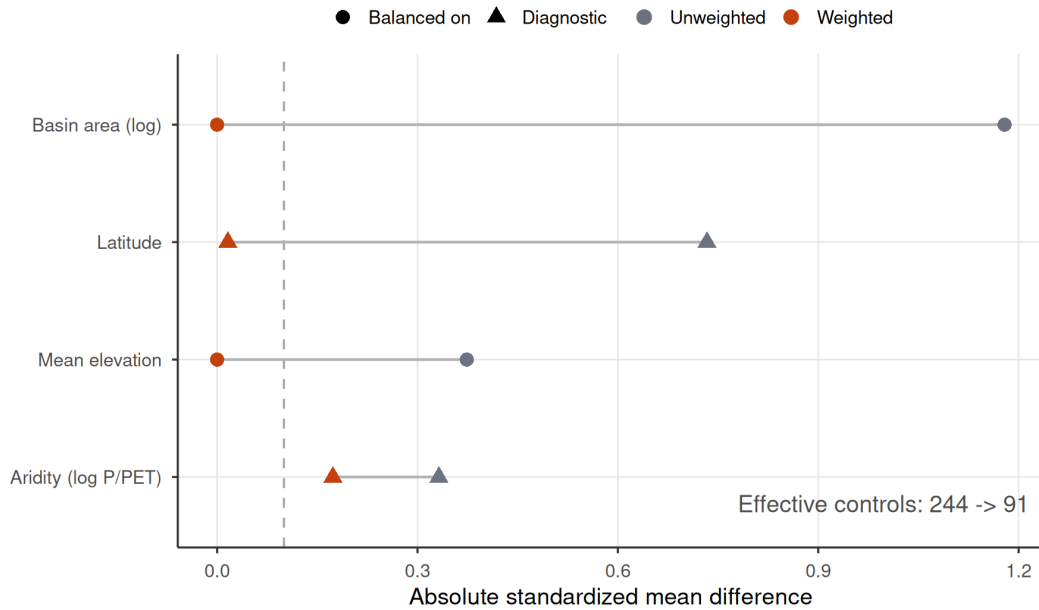
### Supplementary Figure S3 | The national matched design across Chile's aridity gradient.

(a) Matched sample of subcuencas, each filled by long-term aridity (precipitation over potential evapotranspiration,  $P/PET$ , the north-south gradient the design conditions on); dammed (treated) basins are outlined in orange and marked with their reservoir (triangles), and each matched-control basin carries a centroid dot sized by its entropy-balancing weight, so the effective control sample (91) is visible against the nominal 244. Dammed basins concentrate in the arid centre and north; weighted controls span the same climate strata. (b) Within-basin placebo geometry for one representative dammed basin: the reservoir (triangle) with its downstream (regulated) gauges below the dam and upstream (unregulated) gauges above it.

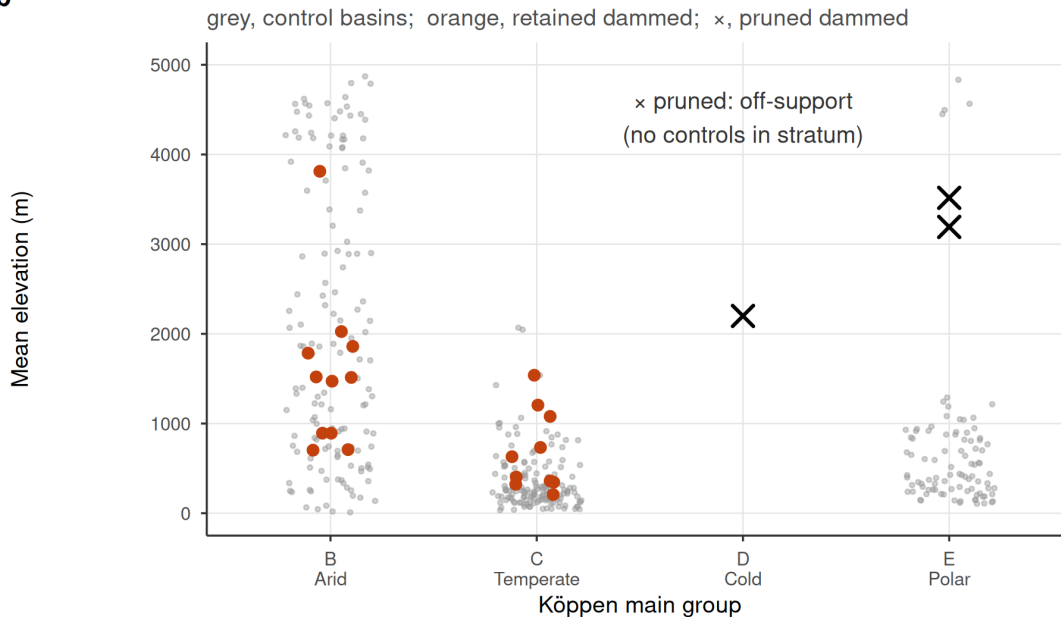
## Matching removes the siting imbalance on common support

Weighting collapses area/elevation gaps to  $\sim 0$ ; pruned units are off-support high-Andes basins

**a**



**b**



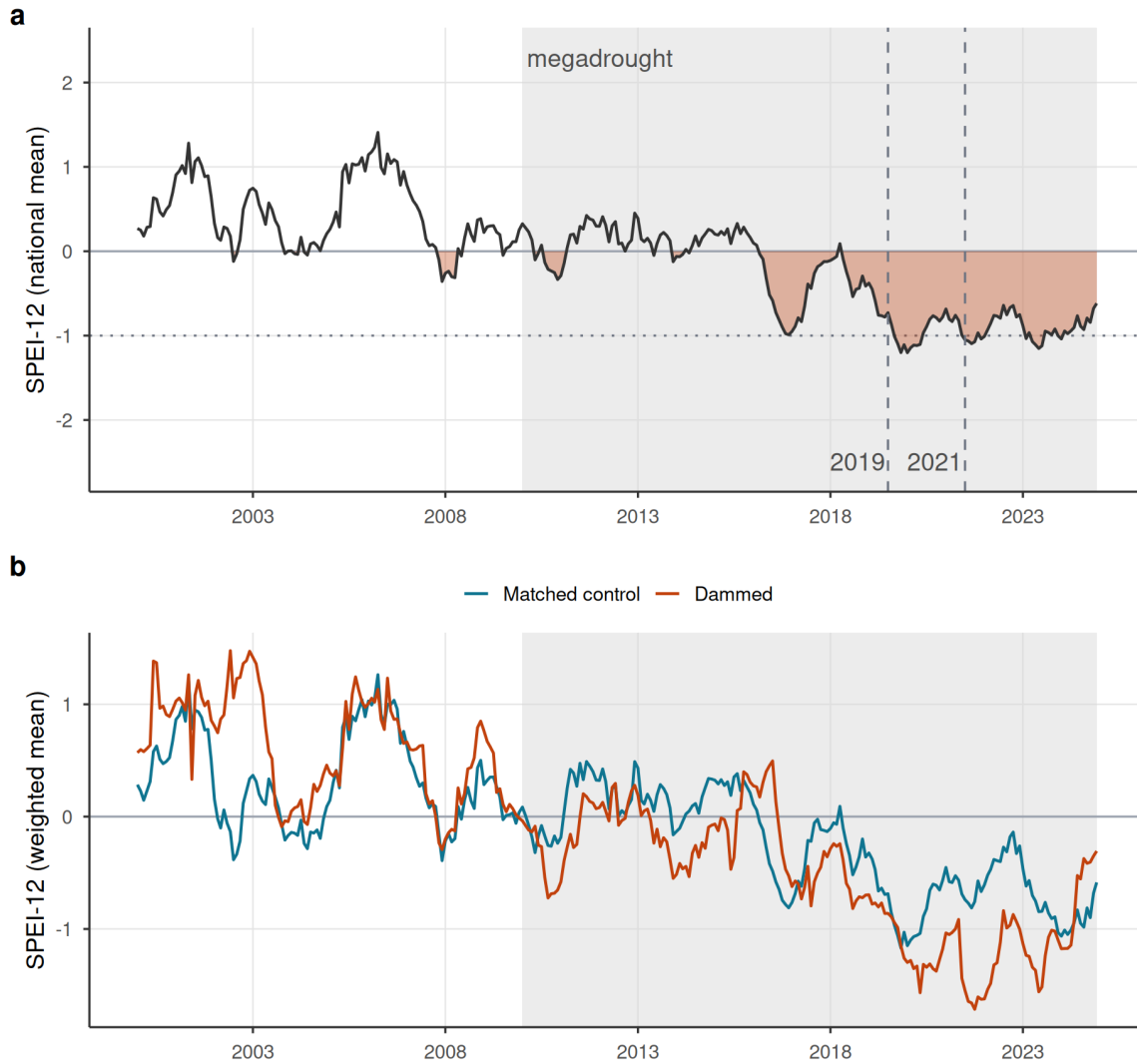
**Supplementary Figure S4 | Covariate balance and common support for the matched design.** (a) Love plot of the absolute standardized mean difference (SMD) between dammed and control basins before (unweighted) and after entropy balancing, for the two balanced covariates (log basin area, mean elevation) and two untargeted diagnostics (latitude, log aridity); the dashed line is the conventional 0.1 balance threshold. Weighting collapses the large siting imbalances to  $\sim 0$ , leaving a small residual aridity difference (0.17) that the doubly-robust outcome model absorbs, and reduces the effective control sample from 244 to 91. (b) Climate-stratum common support: mean elevation of control (grey) and dammed (orange, retained) basins within each Köppen main group;

the three pruned dammed basins (x) fall in the cold (D) and polar (E) strata, whose controls lie far below them, so they are off-support and excluded rather than matched by extrapolation.



### A sustained megadrought forcing, shared by dammed and control basins

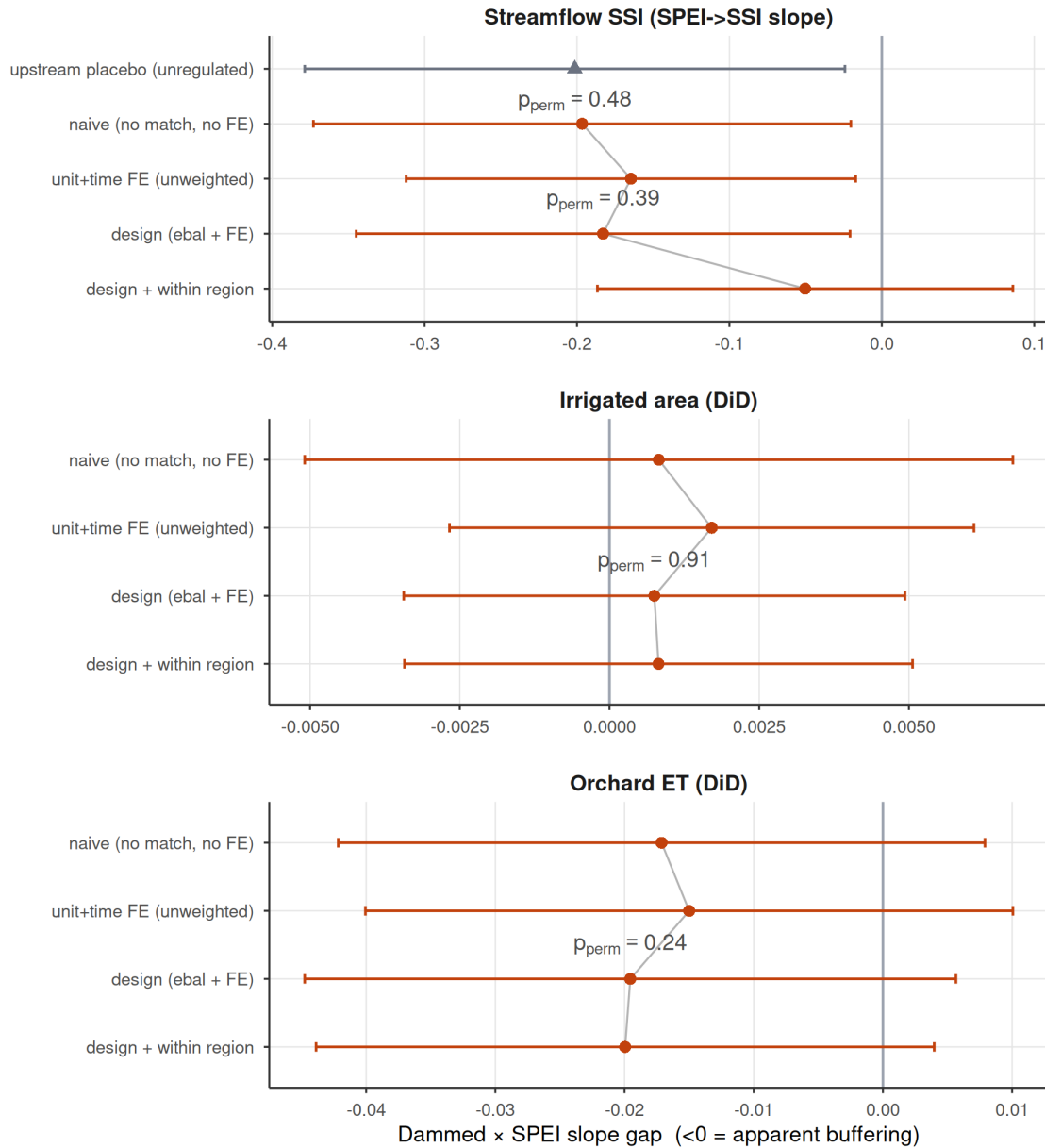
Both track the same shock (dammed run slightly drier, a siting offset), so the estimand is the slope, not level



**Supplementary Figure S5 | The drought forcing.** (a) National mean SPEI-12 across the matched basins, 2000–2024; the negative (drought) excursion is filled and the post-2010 megadrought era shaded, with the 2019 and 2021 hyperdrought winters marked. The deficit is sustained and deep (below  $-1$  for much of 2016–2024), giving the SPEI dose real dynamic range. (b) Entropy-balancing-weighted mean SPEI-12 for dammed versus matched-control basins: both co-move through the same megadrought (a common shock that year fixed effects absorb), with dammed basins running slightly drier, a siting offset that is precisely why the estimand is the differential deficit-to-impact slope rather than a level or calendar-time contrast.

### Apparent buffering is a between-region siting confound

Survives matching and fixed effects; collapses only within climate region

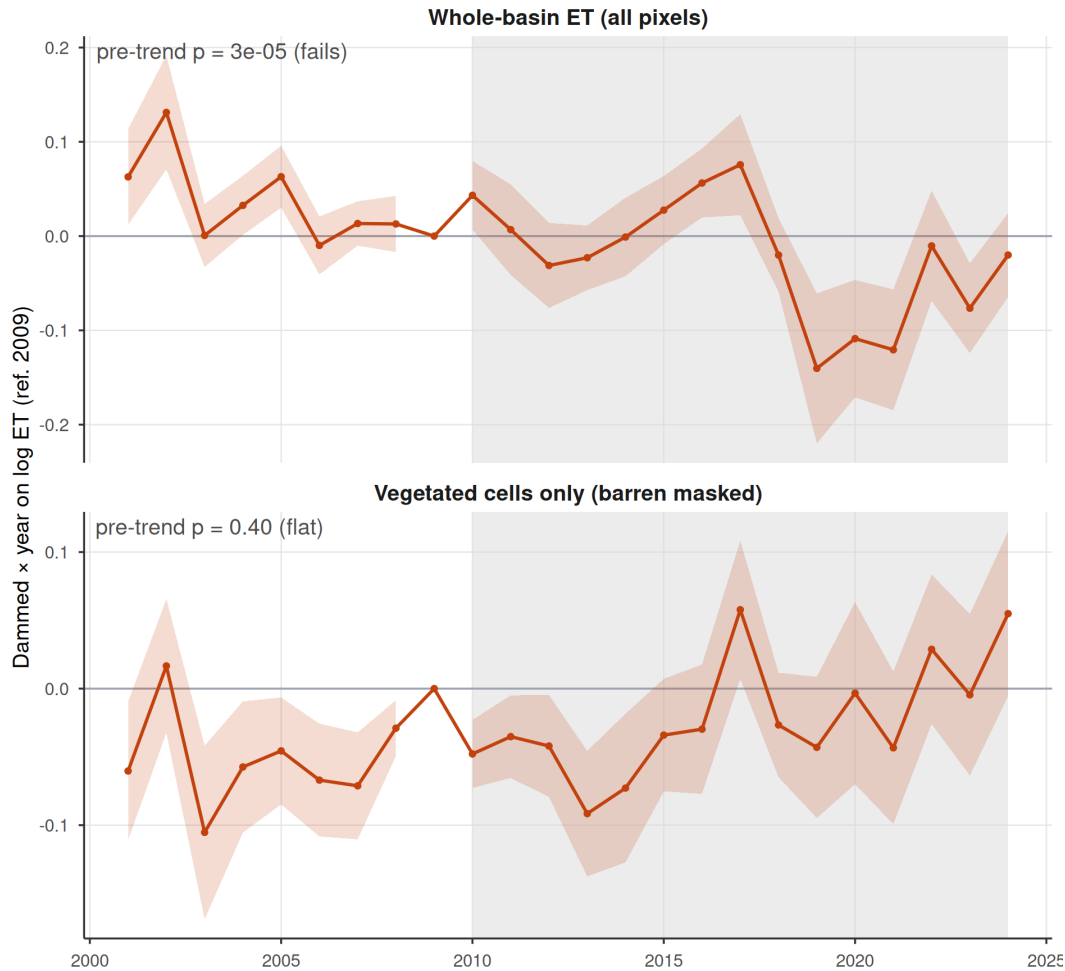


**Supplementary Figure S6 | Siting-confound decomposition ladder.** Dammed-versus-control transmission-slope gap (treat:SPEI; <0 = apparent buffering) estimated by increasingly stringent estimators, for the streamflow SSI slope (top) and the two difference-in-differences proxy outcomes; bars are 95% cluster-robust confidence intervals and the design and within-region rows carry the design's randomization-inference p. Rungs: (1) naive pooled (no matching, no fixed effects, what a conventional dammed-versus-undammed study reports); (2) unit and time fixed effects, unweighted; (3) the design estimator (entropy-balance weights + fixed effects); (4) design with the baseline SPEI-to-outcome slope allowed to vary within climate region. For streamflow the apparent buffering survives rungs (1) to (3) near  $-0.18$  (permutation-null throughout), then collapses toward zero at rung (4), and is equally strong at the unregulated upstream placebo (grey triangle): the signature of a between-region aridity confound, not regulation. The proxy outcomes carry no signal at any

rung.

### The positive whole-basin ET effect is a barren-land artifact

Masking barren pixels (vegetated cells only) removes the pre-trend failure and the effect



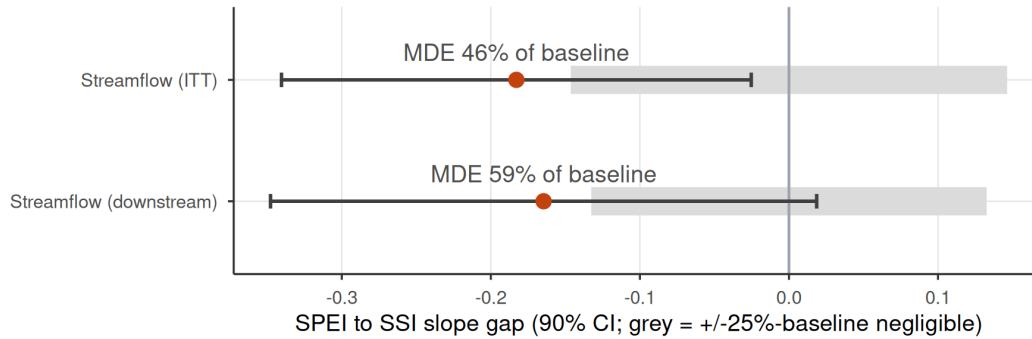
### Supplementary Figure S7 | The positive whole-basin ET effect is a barren-land artifact.

Event studies of the dammed-versus-control evapotranspiration response (dammed x year, ref. 2009) for (a) whole-basin ET over all pixels, which shows strongly non-parallel pre-trends (pre-trend slope  $p = 3 \times 10^{-5}$ ) and an apparent post-2010 divergence, and (b) ET restricted to vegetated irrigated (orchard) cells, i.e. with barren and salar pixels masked out. Masking the barren land removes both the pre-trend failure ( $p = 0.40$ ) and the positive effect (the forcing-conditioned slope gap moves from +0.048 to -0.020), confirming that the exclusion of whole-basin ET from the main results is mechanistic (MOD16 ET is contaminated over sparsely vegetated barren surfaces), not a post-hoc drop of an inconvenient result.

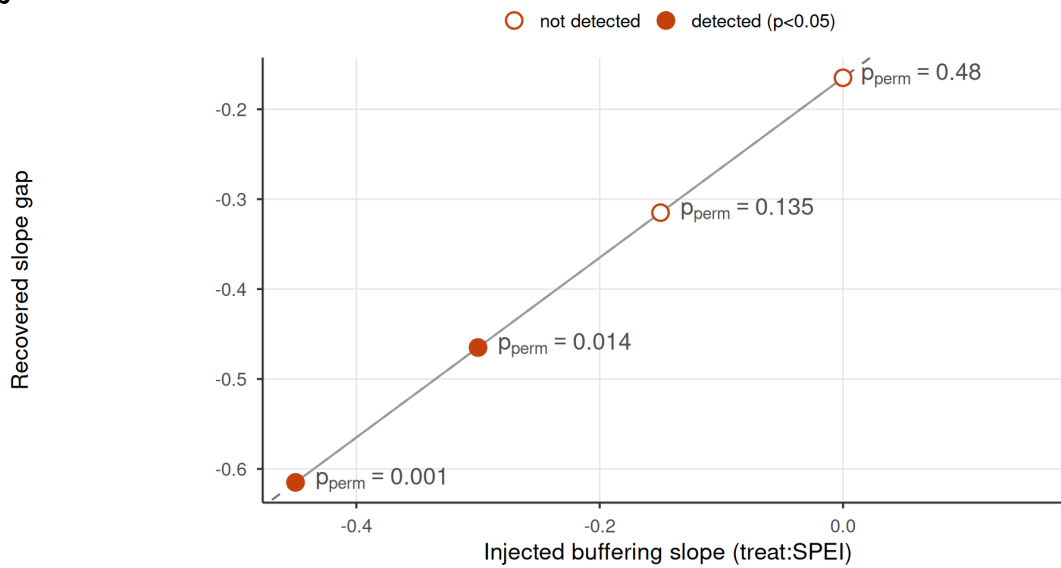
## The null is informative: large effects excluded, injected effects recovered

Rules out buffering beyond ~half the baseline slope; recovers a real one 1:1 (detected at  $|\beta| \geq 0.30$ )

**a**



**b**



**Supplementary Figure S8 | The null is informative, not underpowered blindness.** (a) Equivalence / minimum-detectable-effect bounds for the two streamflow outcomes: the observed slope gap (point) and its 90% two one-sided test (TOST) interval against the  $\pm 25\%$ -of-baseline negligible region (grey). The 90% intervals exceed the negligible region, so formal equivalence to zero is not claimed, but they exclude attenuations larger than the minimum detectable effect (46% of baseline transmission for the intent-to-treat design, 59% downstream). (b) Positive control: a known buffering slope injected into the treated downstream gauges is recovered one-to-one by the estimator (points on the dashed expected-recovery line), the no-injection case is correctly not flagged, and randomization inference detects the effect (filled points) once the injected slope reaches  $-0.30$ . The estimator would therefore reveal a large operational buffering effect if one existed.

## Supplementary Tables

### Supplementary Table S1 | Reservoir-use heterogeneity: irrigation-only robustness.

The treated sample is dominated by irrigation reservoirs (18 of the 21 matched treated basins hold one; three are purely hydropower or potable supply). The headline streamflow SPEI-to-SSI slope gap ( $\text{treat:SPEI}$ ,  $< 0$  = apparent buffering) is re-estimated on all treated basins and on the irrigation-only subset (non-irrigation treated basins dropped, controls unchanged), with the design's stratified randomization-inference p. The slope gap and its permutation-null status are unchanged, so the null is not an artefact of pooling hydropower with irrigation reservoirs; **n\_treat** columns give the treated basins with streamflow gauges in each case.

| Quantity                                   | All<br>treated | p<br>(all) | Irrigation<br>only | p<br>(irrig) | n_treat<br>(all) | n_treat<br>(irrig) |
|--------------------------------------------|----------------|------------|--------------------|--------------|------------------|--------------------|
| Streamflow slope gap (ITT)                 | -0.183         | 0.388      | -0.192             | 0.432        | 17               | 16                 |
| Streamflow slope gap<br>(upstream placebo) | -0.201         | 0.482      | -0.221             | 0.456        | 15               | 14                 |

**Supplementary Table S2 | Inter-basin spillover and administrative-demand checks.** Top rows: matched treated and effective-control basins occupy entirely disjoint DGA cuencas, so no matched pair shares a river network and the within-cuenca contamination rule already removes the direct transfer pathway. The residual transfer bias is bounded in both directions: a transfer moving water from a dammed basin to a control raises the control’s availability and shrinks the estimated gap, biasing toward the null we report, while imports sustaining a treated basin during peak drought would flatten the treated transmission slope where deliveries land, downstream, and so mimic regulation-specific buffering; the absence of any downstream-specific flattening in the placebo bounds this direction too, though a vulnerability signal partially offset by imports cannot be fully excluded without an explicit transfer-network model. Bottom rows: the water-rights registry was not closed during the megadrought, and the closure that would matter, a freeze specific to dammed basins, is absent: treated basins entered the drought with a higher allocated stock per unit area than their weighted controls, accrued new rights about 2.7 times faster through the megadrought, and none went without new grants, while 29% of weighted controls did. The accrual null is therefore a genuine null on new state grants, not the forced silence of saturated registries; what administrative closure does imply is that grant accrual under-represents total demand, because reallocation in closed basins proceeds through private market transfers the grant registry does not record (main text).

| Metric                                                                                   | Value   | Detail                                                                                                   |
|------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------|
| treated cuencas                                                                          | 14.00   | distinct DGA cuencas holding $\geq 1$ treated subcuenca                                                  |
| effective-control cuencas                                                                | 70.00   | distinct DGA cuencas holding $\geq 1$ effective (w>0) control                                            |
| cuencas shared by a matched pair                                                         | 0.00    | contamination rule removes same-cuenca controls $\Rightarrow$ no matched pair shares a river network     |
| new consumptive rights per megadrought year                                              | 4687.00 | range 2539-7352 over 2010-2021; registry not closed                                                      |
| basins granted new rights per year                                                       | 194.00  | of 265 matched basins                                                                                    |
| pre-drought (2009) rights stock, treated (per 100 km <sup>2</sup> , weighted)            | 15.50   | control 6.2; higher stock over lower renewable supply $\Rightarrow$ treated utilization at least as high |
| pre-drought (2009) winsorized volume stock, treated (1/s per km <sup>2</sup> , weighted) | 1.84    | control 0.65                                                                                             |
| megadrought accrual rate, treated (rights per 100 km <sup>2</sup> per yr, weighted)      | 1.72    | control 0.63; treated accrual not slower $\Rightarrow$ no differential administrative closure            |
| treated basins with zero new rights over the megadrought (share)                         | 0.00    | control share 0.29                                                                                       |

**Supplementary Table S3 | Control screening against the national dam inventory.**

“Undammed” is defined against the 26 monitored reservoirs, so the matched controls are screened against the full 1,370-impoundment inventory: 37 of 244 matched control basins contain some impoundment, 10 contain a material dam (monitored, registry-large, or wall at least 15 m), and those 10 carry 5.5% of the entropy-balancing weight. Re-fitting with the entropy balance re-estimated on the dam-free control pool leaves every streamflow estimate unchanged, so control-side contamination by unmonitored dams does not attenuate the estimates toward zero.

| Quantity                                            | Value Detail                                                                                                                                    |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| matched control basins containing any inventory dam | 37.000 of 244 matched controls; 10 contain a MATERIAL dam (monitored / GRANDE / wall $\geq 15$ m), holding 5.5% of the entropy-balancing weight |
| dam-free-control buffering, ITT (re-balanced)       | - perm p = 0.353 (1000 perms); primary -0.183; entropy balance 0.183 re-fit on dam-free controls, control ESS 79.6                              |
| dam-free-control upstream placebo (re-balanced)     | - perm p = 0.472; primary -0.201<br>0.202                                                                                                       |
| dam-free-control placebo contrast (down - up)       | 0.037 down -0.165, up -0.202, perm p = 0.700 (1000 perms)                                                                                       |



#### Supplementary Table S4 | Capacity-weighted and absolute-volume storage-band trends.

The main-text band trends pool reservoirs with equal weight although capacities span three orders of magnitude, so a steep decline in small reservoirs could in principle dominate the pooled percent-of-capacity trend while the large reservoirs holding most of the national buffer stayed stable. Re-fitting with capacity weights shows the opposite: the weighted decline is steeper than the unweighted one, the peak stabilizes after 2010 while the trough keeps falling, and the weighted seasonal amplitude widens ( $+0.8$  pp yr<sup>-1</sup>); the flat unweighted amplitude and the widening weighted amplitude both run opposite to the narrowing that sedimentation or a failing refill cycle would produce. The p-values here are analytic and cluster-robust; because capacity weighting concentrates effective weight on a few large reservoirs and so sharpens the small-cluster problem, the governing inference for these coefficients is the wild cluster bootstrap of Supplementary Table S6, under which the weighted trough trend and drift and the amplitude widening survive while the weighted peak trend and the weighted 2010 steps do not. In absolute volume, the coverage-stable subset (22 reservoirs, 98% of fleet capacity) loses 88 hm<sup>3</sup> yr<sup>-1</sup> of peak and 209 hm<sup>3</sup> yr<sup>-1</sup> of trough storage over 2005–2024. The registry capacities are static nominal values; the sedimentation reading of the percent-of-capacity decline is bounded in the main text (step timing, unchanged amplitude, co-movement with unregulated control streamflow).

| Quantity                                                                       | Value    | Detail                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| capacity-weighted trend (per yr), peak                                         | - 0.0087 | SE 0.0030, p = 0.007 (capacity-weighted reservoir FE, clustered)                                                                                                                                        |
| capacity-weighted trend (per yr), trough                                       | 0.0166   | - SE 0.0017, p = 5.7e-10 (capacity-weighted reservoir FE, clustered)                                                                                                                                    |
| capacity-weighted trend (per yr), amplitude                                    | 0.0079   | SE 0.0022, p = 0.00124 (capacity-weighted reservoir FE, clustered)                                                                                                                                      |
| capacity-weighted step at 2010, peak                                           | 0.1781   | - SE 0.0515, p = 0.00205 (capacity-weighted reservoir FE, clustered)                                                                                                                                    |
| capacity-weighted step at 2010, trough                                         | 0.1165   | - SE 0.0554, p = 0.0462 (capacity-weighted reservoir FE, clustered)                                                                                                                                     |
| capacity-weighted post-2010 drift (per yr), peak                               | 0.0009   | SE 0.0011, p = 0.43 (capacity-weighted reservoir FE, clustered)                                                                                                                                         |
| capacity-weighted post-2010 drift (per yr), trough                             | 0.0127   | - SE 0.0028, p = 0.000143 (capacity-weighted reservoir FE, clustered)                                                                                                                                   |
| coverage-stable subset share of fleet capacity                                 | 0.9770   | 22 of 25 reservoirs with $\geq 18$ band years; totals below use years with all 22 reporting                                                                                                             |
| fleet total peak storage (hm <sup>3</sup> ), trend and 2010 step               | 88.0000  | - trend -88.0 hm <sup>3</sup> /yr (p = 0.0216); 2010 step -1549 hm <sup>3</sup> (p = 0.106), post-2010 drift +204.9 hm <sup>3</sup> /yr (p = 0.518); 22 reservoirs with full coverage, 20 of 21 years   |
| fleet total trough (carryover) storage (hm <sup>3</sup> ), trend and 2010 step | 209.2000 | - trend -209.2 hm <sup>3</sup> /yr (p = 7.53e-05); 2010 step -1254 hm <sup>3</sup> (p = 0.384), post-2010 drift -71.3 hm <sup>3</sup> /yr (p = 0.884); 22 reservoirs with full coverage, 20 of 21 years |

**Supplementary Table S5 | Storage-band trends on the coverage-stable subset.** Several reservoirs enter the monitored panel after 2005, and their filling phases are not absorbed by static reservoir fixed effects. Restricting to the 22 reservoirs observed in at least 18 of the 20 band years reproduces the full-fleet pattern: peak and trough decline, no amplitude narrowing, the peak steps down at 2010 with no further drift, and the trough keeps eroding, so mid-panel commissioning does not tilt the trends.

| Quantity                                       | Value    | Detail                                                                                                          |
|------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------|
| stable-subset trend (per yr), peak             | - 0.0134 | SE 0.0050, $p = 0.0144$ (22 reservoirs with $\geq 18$ band years; full fleet in Supplementary Tables S5, S41)   |
| stable-subset trend (per yr), trough           | - 0.0122 | SE 0.0027, $p = 0.000177$ (22 reservoirs with $\geq 18$ band years; full fleet in Supplementary Tables S5, S41) |
| stable-subset trend (per yr), amplitude        | - 0.0012 | SE 0.0057, $p = 0.832$ (22 reservoirs with $\geq 18$ band years; full fleet in Supplementary Tables S5, S41)    |
| stable-subset step at 2010, peak               | - 0.1935 | SE 0.0907, $p = 0.0449$ (22 reservoirs with $\geq 18$ band years; full fleet in Supplementary Tables S5, S41)   |
| stable-subset step at 2010, trough             | - 0.0743 | SE 0.0475, $p = 0.133$ (22 reservoirs with $\geq 18$ band years; full fleet in Supplementary Tables S5, S41)    |
| stable-subset post-2010 drift (per yr), peak   | - 0.0024 | SE 0.0038, $p = 0.532$ (22 reservoirs with $\geq 18$ band years; full fleet in Supplementary Tables S5, S41)    |
| stable-subset post-2010 drift (per yr), trough | - 0.0100 | SE 0.0037, $p = 0.014$ (22 reservoirs with $\geq 18$ band years; full fleet in Supplementary Tables S5, S41)    |

**Supplementary Table S6 | Wild-cluster-bootstrap inference for the storage-band models.**

The storage trends, the 2010 steps, and the post-2010 drifts are estimated with 22 to 26 reservoir clusters, where cluster-robust standard errors can over-reject; a descriptive within-reservoir trend offers no treatment assignment for the design’s randomization inference to permute, so the small-cluster refinement is the restricted (null-imposed) wild cluster bootstrap with Rademacher weights drawn at the reservoir level (2,999 draws per coefficient). Bootstrap p-values are reported beside the analytic ones for every coefficient, unweighted and capacity-weighted. The unweighted results, which carry the main-text claims, tighten under the bootstrap: the peak trend moves from  $p = 0.013$  to 0.004 and the 2010 peak step from 0.04 to 0.009, while the amplitude trend and the trough step stay null. The capacity-weighted fits behave oppositely, because weighting concentrates effective weight on a few large reservoirs and so sharpens the small-cluster problem: the weighted trough trend and drift survive ( $p = 0.001$ ) and the weighted amplitude widening survives ( $p = 0.031$ ), but the weighted peak trend ( $p = 0.13$ ) and both weighted 2010 steps ( $p = 0.088$  and 0.28) do not, so the analytic p-values reported in Supplementary Table S4 for those three coefficients over-reject and the step evidence rests on the unweighted fit.

| Quantity                          | Value    | Detail                                                                                                             |
|-----------------------------------|----------|--------------------------------------------------------------------------------------------------------------------|
| unweighted trend (per yr), peak   | - 0.0128 | analytic cluster-robust $p = 0.0127$ ; wild cluster bootstrap $p = 0.004$ (1999 draws, Rademacher, null imposed)   |
| unweighted trend (per yr), trough | - 0.0122 | analytic cluster-robust $p = 7.33e-05$ ; wild cluster bootstrap $p = 0.001$ (1999 draws, Rademacher, null imposed) |

| Quantity                                           | Value  | Detail                                                                                                               |
|----------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------|
| unweighted trend (per yr), amplitude               | 0.0006 | - analytic cluster-robust $p = 0.912$ ; wild cluster bootstrap $p = 0.984$ (1999 draws, Rademacher, null imposed)    |
| unweighted step at 2010, peak                      | 0.1915 | - analytic cluster-robust $p = 0.04$ ; wild cluster bootstrap $p = 0.009$ (1999 draws, Rademacher, null imposed)     |
| unweighted step at 2010, trough                    | 0.0741 | - analytic cluster-robust $p = 0.118$ ; wild cluster bootstrap $p = 0.105$ (1999 draws, Rademacher, null imposed)    |
| unweighted post-2010 drift (per yr), peak          | 0.0024 | - analytic cluster-robust $p = 0.487$ ; wild cluster bootstrap $p = 0.513$ (1999 draws, Rademacher, null imposed)    |
| unweighted post-2010 drift (per yr), trough        | 0.0102 | - analytic cluster-robust $p = 0.00653$ ; wild cluster bootstrap $p = 0.009$ (1999 draws, Rademacher, null imposed)  |
| capacity-weighted trend (per yr), peak             | 0.0087 | - analytic cluster-robust $p = 0.007$ ; wild cluster bootstrap $p = 0.132$ (1999 draws, Rademacher, null imposed)    |
| capacity-weighted trend (per yr), trough           | 0.0166 | - analytic cluster-robust $p = 5.7e-10$ ; wild cluster bootstrap $p = 0.001$ (1999 draws, Rademacher, null imposed)  |
| capacity-weighted trend (per yr), amplitude        | 0.0079 | analytic cluster-robust $p = 0.00124$ ; wild cluster bootstrap $p = 0.031$ (1999 draws, Rademacher, null imposed)    |
| capacity-weighted step at 2010, peak               | 0.1781 | - analytic cluster-robust $p = 0.00205$ ; wild cluster bootstrap $p = 0.088$ (1999 draws, Rademacher, null imposed)  |
| capacity-weighted step at 2010, trough             | 0.1165 | - analytic cluster-robust $p = 0.0462$ ; wild cluster bootstrap $p = 0.276$ (1999 draws, Rademacher, null imposed)   |
| capacity-weighted post-2010 drift (per yr), peak   | 0.0009 | analytic cluster-robust $p = 0.43$ ; wild cluster bootstrap $p = 0.594$ (1999 draws, Rademacher, null imposed)       |
| capacity-weighted post-2010 drift (per yr), trough | 0.0127 | - analytic cluster-robust $p = 0.000143$ ; wild cluster bootstrap $p = 0.001$ (1999 draws, Rademacher, null imposed) |

**Supplementary Table S7 | Carryover-ratio denominator sensitivity.** The carryover ratio (capacity / mean annual flow) uses regulated downstream flow, which consumption can depress; recomputed with the unregulated upstream-gauge flow, the available inflow proxy, the classification is stable for all seven carryover basins and moves one seasonal basin (Corrales, whose upstream gauge drains only a small headwater) into the carryover group. The heterogeneity result keeps its direction under the upflow-based split (carryover  $D < 0$ , seasonal  $D = 0$ ) but the carryover contrast weakens to  $D = -0.165$  ( $p = 0.146$ ) because the added basin does not buffer; the direction is robust to the ratio’s denominator, the significance is not.

| Unit | Reservoir   | Capacity (hm3) | Down-flow ratio | Up-flow ratio | Down-flow class | Up-flow class | Flip |
|------|-------------|----------------|-----------------|---------------|-----------------|---------------|------|
| 210  | Conchi      | 22             | 0.732           | 0.899         | carryover       | carryover     |      |
| 343  | Lautaro     | 16             | 0.314           | 0.311         | seasonal        | seasonal      |      |
| 382  | Santa Juana | 176            | 1.160           | 1.403         | carryover       | carryover     |      |
| 432  | Puclaro     | 209            | 0.718           | 0.617         | carryover       | carryover     |      |
| 450  | Recoleta    | 100            | 2.218           | 1.075         | carryover       | carryover     |      |
| 453  | Cogotí      | 156            | 2.546           | 3.176         | carryover       | carryover     |      |
| 471  | Corrales    | 50             | 0.148           | 2.002         | seasonal        | carryover     | yes  |
| 472  | El Bato     | 26             | 0.294           | 0.356         | seasonal        | seasonal      |      |

| Unit | Reservoir      | Capacity<br>(hm3) | Down-flow<br>ratio | Up-flow<br>ratio | Down-flow<br>class | Up-flow<br>class | Flip |
|------|----------------|-------------------|--------------------|------------------|--------------------|------------------|------|
| 603  | Convento Viejo | 237               | 0.193              | 0.484            | seasonal           | seasonal         |      |
| 732  | Colbún         | 1544              | 0.658              | 0.511            | carryover          | carryover        |      |
| 733  | Digua          | 225               | 0.244              | 0.300            | seasonal           | seasonal         |      |
| 735  | Bullileo       | 60                | 0.033              | NA               | seasonal           | no-up-gauge      | yes  |
| 811  | Coihueco       | 30                | 0.039              | 0.156            | seasonal           | seasonal         |      |
| 831  | Ralco          | 1174              | 0.180              | 0.380            | seasonal           | seasonal         |      |
| 837  | Lago Laja      | 5582              | 7.522              | 28.619           | carryover          | carryover        |      |

| Subset                                        | n treat | ITT    | ITT<br>p | Down   | Up     | D      | Perm.<br>p |
|-----------------------------------------------|---------|--------|----------|--------|--------|--------|------------|
| carryover reservoirs (capacity $\geq 0.5$ yr) | 8       | -0.293 | 0.264    | -0.383 | -0.218 | -0.165 | 0.146      |
| seasonal reservoirs (capacity $< 0.5$ yr)     | 9       | -0.077 | 0.584    | 0.004  | -0.184 | 0.188  | 0.272      |

**Supplementary Table S8 | Strictly-paired carryover contrast.** The capacity split (7 carryover + 10 seasonal = 17) is on the buffering sample, the treated basins with a usable downstream gauge; the down/up contrast within a subgroup uses each subgroup's available panels, which are only partly paired. Restricting both sides of the carryover contrast to the six basins with usable gauges in both classes leaves it essentially unchanged ( $D = -0.234$ , permutation  $p = 0.004$ , versus  $D = -0.232$ ,  $p = 0.014$  in the full split), so the carryover buffering does not rest on the partially-paired composition.

| Subset                          | n paired | D      | Perm. p | Down   | Up     | n perms |
|---------------------------------|----------|--------|---------|--------|--------|---------|
| strictly-paired carryover (n=6) | 6        | -0.234 | 0.004   | -0.459 | -0.225 | 1000    |

**Supplementary Table S9 | Carryover-claim hardening: hydropower exclusion and leave-one-out inference.** The carryover class is not a random draw of the fleet: it mixes five irrigation reservoirs (Conchi, Santa Juana, Puclaro, Recoleta, Cogotí) with two large dams whose operation involves hydropower generation (Colbún, registry use irrigation/generation; Lago Laja, operated jointly for irrigation and generation), whose scheduling could smooth flows for reasons unrelated to drought carryover. The first row re-estimates the carryover contrast on the irrigation-only carryover basins (controls unchanged). The remaining rows re-run the contrast dropping one carryover basin at a time, each with its own permutation null, so the stability of the p-value, not only of the point estimate, is visible; with seven treated units a single influential basin could otherwise move the permutation p across conventional thresholds.

| Subset                                               | n<br>treat | ITT |       |       |       |       | Perm.<br>p |
|------------------------------------------------------|------------|-----|-------|-------|-------|-------|------------|
|                                                      |            | ITT | p     | Down  | Up    | D     |            |
| irrigation-only carryover (hydropower excluded, n=5) | 5          | -   | 0.246 | -     | -     | -     | 0.057      |
| leave-one-out: drop Conchi                           | 6          | -   | 0.254 | -     | -     | -     | 0.032      |
|                                                      |            |     | 0.355 | 0.478 | 0.249 | 0.229 |            |
| leave-one-out: drop Santa Juana                      | 6          | -   | 0.317 | -     | -     | -     | 0.066      |
|                                                      |            |     | 0.258 | 0.401 | 0.175 | 0.226 |            |
| leave-one-out: drop Puclaro                          | 6          | -   | 0.336 | -     | -     | -     | 0.020      |
|                                                      |            |     | 0.281 | 0.397 | 0.165 | 0.232 |            |
| leave-one-out: drop Recoleta                         | 6          | -   | 0.270 | -     | -     | -     | 0.004      |
|                                                      |            |     | 0.344 | 0.459 | 0.225 | 0.234 |            |
| leave-one-out: drop Cogotí                           | 6          | -   | 0.278 | -     | -     | -     | 0.050      |
|                                                      |            |     | 0.340 | 0.470 | 0.237 | 0.233 |            |
| leave-one-out: drop Colbún                           | 6          | -   | 0.283 | -     | -     | -     | 0.022      |
|                                                      |            |     | 0.367 | 0.549 | 0.266 | 0.283 |            |
| leave-one-out: drop Lago Laja                        | 6          | -   | 0.264 | -     | -     | -     | 0.120      |
|                                                      |            |     | 0.342 | 0.488 | 0.269 | 0.219 |            |

**Supplementary Table S10 | Drought-conditional streamflow missingness.** The no-imputation rule deletes months rather than fills them, so its selection structure is quantified directly. Missingness is genuinely drought-selective (33.8% of drought gauge-months missing against 13.9% otherwise), a truncation of the dry tail shared by treated and control gauges alike, which the differential estimand nets out; what could bias the placebo contrast is a *differential* between gauge classes, and that differential is small and slightly favors upstream loss (32.9% downstream against 35.1% upstream in drought months), the direction that would, if anything, thin the placebo rather than protect it. The dry-tail re-estimates (Supplementary Table S40) and the long-baseline SSI (Supplementary Table S11) carry the same verdict on the retained months.

| Quantity                                                         | Value         | Detail                                                                                                          |
|------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------|
| gauge-month missingness within record span, 2000-2020            | 0.1567015.67% | of 47124 gauge-months (months failing the $\geq 20$ -valid-days rule or absent)                                 |
| missingness in drought months (SPEI $< -1$ ) vs other months     | 0.33790       | drought 33.79% vs non-drought 13.90%                                                                            |
| missingness on SPEI (gauge FE, unit-clustered)                   | 0.05068       | - per SPEI unit; SE 0.00891, $p = 2.27\text{e-}07$ ; a negative sign means drier months are MORE likely missing |
| drought-month missingness, treated downstream vs upstream gauges | 0.02160       | - down 32.92%, up 35.09%; a differential would be needed to bias the contrast D                                 |

**Supplementary Table S11 | SSI standardized on the 1990–2020 baseline.** The primary SSI uses the common 2000–2020 window so every gauge shares one reference period; that window is megadrought-dominated and shorter than the 30-year guideline. Re-standardizing on 1990–2020 (31 years; the CR2 daily record, from the Center for Climate and Resilience Research, begins in 1900) leaves the estimation window unchanged and extends only the standardization reference: every buffering estimate and the placebo contrast are essentially unchanged, so the transmission slopes are not an artifact of a drought-heavy reference period.

| Quantity                                        | Value   | Detail                                                                      |
|-------------------------------------------------|---------|-----------------------------------------------------------------------------|
| gauges retained under 1990-2020 standardization | 171.000 | SSI-12 re-standardized per gauge-month (Gringorten) on the 31-year baseline |
| long-baseline buffering, ITT                    | - 0.188 | perm p = 0.280 (1000 perms); 2000-2020-baseline primary -0.183              |
| long-baseline buffering, downstream             | - 0.178 | perm p = 0.284 (1000 perms); 2000-2020-baseline primary -0.165              |
| long-baseline buffering, upstream placebo       | - 0.198 | perm p = 0.458 (1000 perms); 2000-2020-baseline primary -0.201              |
| long-baseline placebo contrast (down - up)      | 0.020   | perm p = 0.893 (1000 perms); regulation would give < 0                      |

**Supplementary Table S12 | Volumetric-scale streamflow checks.** SSI standardizes each gauge on its own, for regulated gauges post-dam, flow distribution, so the transmission slope tests distribution-relative propagation and would divide out any dam-induced compression of flow variance. Two direct checks show the streamflow null is not a standardization artifact. First, the buffering models and the upstream placebo are re-estimated on log 12-month mean flow, a volumetric semi-elasticity (percent of flow per SPEI unit) that no per-gauge normalization can compress: all slope gaps remain permutation-null, with point estimates slightly positive (toward steeper, not buffered, transmission) and the upstream placebo at least as large as downstream, reproducing the siting pattern. Second, the compression premise itself: deseasonalized log-flow variability at regulated downstream gauges is statistically indistinguishable from unregulated upstream gauges (Wilcoxon p = 0.43), so there is no differential variance compression for the standardization to divide out at the monthly-to-annual scale our design resolves.

| Quantity                                                         | Value                                                                                                      | Detail                                                                 |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| log-flow semi-elasticity gap, ITT (treat x SPEI)                 | 0.0743SE 0.0720                                                                                            | perm p = 0.562; baseline d log Q / d SPEI 0.139 (% flow per SPEI unit) |
| log-flow semi-elasticity gap, downstream (treat x SPEI)          | 0.0401SE 0.0855                                                                                            | perm p = 0.782; baseline d log Q / d SPEI 0.099 (% flow per SPEI unit) |
| log-flow semi-elasticity gap, upstream placebo (treat x SPEI)    | 0.0952SE 0.0682                                                                                            | perm p = 0.422; baseline d log Q / d SPEI 0.168 (% flow per SPEI unit) |
| deseasonalized log-flow SD by gauge class (relative variability) | 0.8270downstream 0.827 (n=61) vs upstream 0.902 (n=35) vs control 0.579 (n=121); up-down Wilcoxon p = 0.43 |                                                                        |



**Supplementary Table S13 | Sensitivity of the water-rights volume null to the winsorization threshold.** The winsorized allocated-volume expansion ATT and its randomization-inference p, with individual consumptive flows capped at the 95th, 99th, 99.5th (the value used in the main text) and 99.9th percentiles of the consumptive-flow distribution. The analytic interval excludes zero at every threshold, but the effect is not significant under randomization inference at any threshold (permutation p from 0.39 to 0.21), so the volume null is not an artefact of the winsorization choice. The cap is also physically validated (second table): the 99.5th-percentile cap corresponds to  $0.3 \text{ m}^3 \text{ s}^{-1}$ , and in every gauged basin the largest retained right is below the basin’s maximum observed monthly mean streamflow (median ratio 0.002), so no physically impossible values survive the cap; the uncapped registry maximum, by contrast, exceeds  $2 \times 10^8 \text{ l/s}$ .

| Winsor pctl | Volume ATT | 95% CI        | p_perm |
|-------------|------------|---------------|--------|
| 95.0        | 1.837      | [0.639, 3.03] | 0.389  |
| 99.0        | 2.395      | [0.956, 3.83] | 0.343  |
| 99.5        | 2.539      | [1.07, 4.01]  | 0.310  |
| 99.9        | 3.211      | [1.56, 4.86]  | 0.209  |

| Quantity                                                                        | Value       | Detail                                                                                    |
|---------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------|
| winsorization cap (99.5th pct of consumptive flows)                             | 287.000 l/s | (= $0.3 \text{ m}^3/\text{s}$ ); uncapped registry maximum $2.08 \times 10^8 \text{ l/s}$ |
| gauged basins where the largest retained right $\leq$ max observed monthly flow | 77.000      | of 77 gauged basins (100%)                                                                |
| median ratio: largest retained right / max observed flow                        | 0.002       | IQR [0.000, 0.004]                                                                        |

**Supplementary Table S14 | Rank-based tests on raw, uncapped water-rights volumes.** Rights volumes are extremely concentrated and carry known data errors, which is why the level analysis winsorizes; rank-based statistics are immune to caps, outliers, and any monotone error in extreme caudal entries. The per-right volume distribution is nearly identical in dammed and control basins at the median, so the count density does not systematically understate demand in dammed basins. The naive basin-level rank gap in raw added volume reproduces the siting gap, as with the count; the design contrast (ebal-weighted mean-rank difference) appears significant under unit-level permutation, which is anti-conservative given the strong spatial autocorrelation of water-rights outcomes (Supplementary Table S27), and is not significant under the governing cuenca-block permutation, matching the winsorized-volume verdict.

| Quantity                                                                                  | Value | Detail                                                                                          |
|-------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------|
| volume per right, raw (median l/s): dammed vs control                                     | 2.000 | control 1.8; Wilcoxon rank-sum p = 0.00 (n = 77,287 rights)                                     |
| raw volume added 2005-2024, uncapped (median l/s per km <sup>2</sup> ): dammed vs control | 2.310 | control 0.17; naive Wilcoxon p = 0.00 (265 basins) reproduces the siting gap the design removes |

| Quantity                                                                   | Value  | Detail                                                                                                                                            |
|----------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| design rank contrast on raw volume (mean-rank difference, ebal-weighted)   | 84.700 | unit-level stratified permutation $p = 0.001$ , anti-conservative under the spatial autocorrelation of Supp Table S9                              |
| design rank contrast under spatially-restricted (cuenca-block) permutation | 84.700 | cuenca-block permutation $p = 0.145$ (1999 perms), the governing inference for water-rights outcomes; rank scale immune to caudal errors and caps |
| share of total raw volume held by the largest 1% of rights                 | 0.997  | raw registry (data errors included); why the level analysis winsorizes                                                                            |

**Supplementary Table S15 | Snowmelt-confound tests on the upstream/downstream placebo, run on the differential estimand.** Upstream reaches are snow-dominated Andean headwaters, so snowmelt storage and release could in principle shape the treated-versus-control transmission differential up- versus downstream independently of regulation. All snow terms use ERA5-Land snow-water-equivalent (SWE) standardized across units, since the product underestimates absolute Andean snowpack. Four tests bound the concern: a forcing-by-snowpack adjustment leaves the up/down equivalence intact while resolving a detectable cross-sectional snow term upstream, so the product is not blind to snow; the equivalence persists among low-snow units (below-median peak SWE); the apparent buffering does not strengthen in high-snow years (snow-year modulation); and the treated-versus-control differential is statistically indistinguishable between the early and late halves of the panel (period interaction  $p = 0.46$  upstream,  $0.52$  downstream), providing no evidence of a snow buffer degrading in parallel with the megadrought. Gauge-level rows cap any snow-induced up-minus-down slope difference at  $0.007$  against the observed  $0.04$ ; these rows are restricted to placebo-basin gauges with SWE coverage, so gauge counts differ slightly from Supplementary Table S31. Snowmelt is therefore not the driver of the placebo result.

| Quantity                                                                   | ValueDetail                                                                                                                                                                                       |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| upstream placebo: treat x SPEI, unadjusted                                 | - SE 0.0904, $p = 0.030$ ; 15 treated / 44 control units<br>0.2014                                                                                                                                |
| upstream placebo: treat x SPEI, snow-adjusted (spei x peak-SWE z)          | - SE 0.0899, $p = 0.070$ ; snow term 0.0922 (SE 0.0360, $p = 0.165$ )<br>0.013                                                                                                                    |
| upstream placebo: treat x SPEI, low-snow units (below-median peak SWE)     | - SE 0.0791, $p = 0.304$ ; 3 treated / 26 control units; peak SWE<br>0.0829<br>0.0000-0.0012 m w.e.                                                                                               |
| upstream placebo: treat x SPEI, early vs late period                       | - early ( $\leq 2010$ ) -0.3207 (SE 0.1180, $p = 0.009$ ); late ( $> 2010$ )<br>0.3207<br>0.2654 (SE 0.1795, $p = 0.145$ ); period change (treat x SPEI x late) 0.1034 (SE 0.1394, $p = 0.461$ )  |
| upstream placebo: treat x SPEI x annual-SWE anomaly (snow-year modulation) | - SE 0.0472, $p = 0.510$ ; base spei x snow-year 0.0521 (SE 0.0313)<br>0.0374, $p = 0.169$                                                                                                        |
| downstream: treat x SPEI, unadjusted                                       | - SE 0.0851, $p = 0.058$ ; 15 treated / 44 control units<br>0.1646                                                                                                                                |
| downstream: treat x SPEI, snow-adjusted (spei x peak-SWE z)                | - SE 0.0921, $p = 0.131$ ; snow term 0.0296 (SE 0.0262, $p = 0.141$ )<br>0.264                                                                                                                    |
| downstream: treat x SPEI, low-snow units (below-median peak SWE)           | - SE 0.1207, $p = 0.509$ ; 6 treated / 23 control units; peak SWE<br>0.0808<br>0.0000-0.0005 m w.e.                                                                                               |
| downstream: treat x SPEI, early vs late period                             | - early ( $\leq 2010$ ) -0.2060 (SE 0.1276, $p = 0.112$ ); late ( $> 2010$ )<br>0.2060<br>0.3216 (SE 0.1851, $p = 0.088$ ); period change (treat x SPEI x late) -0.1047 (SE 0.1611, $p = 0.518$ ) |
| downstream: treat x SPEI x annual-SWE anomaly (snow-year modulation)       | - SE 0.0809, $p = 0.207$ ; base spei x snow-year 0.0345 (SE 0.1031)<br>0.0328, $p = 0.297$                                                                                                        |

| Quantity                                                  | ValueDetail                                                                             |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------|
| gauges: up vs down peak SWE (m w.e., ERA5-Land)           | 0.0682down 0.0482 (gap 0.0200 m = 0.13 SD); n_up=28, n_down=32                          |
| control gauges: transmission slope per SD peak SWE        | 0.0006SE 0.0282, 95% CI [-0.0553, 0.0553], p = 0.999 (n=82); net of elevation p = 0.808 |
| conservative bound: snow-induced up-down slope difference | 0.0075SWE gap 0.13 SD x CI-upper  sensitivity  0.0553; observed differential gap -0.04  |

**Supplementary Table S16 | Evapotranspiration outcomes (relegated from the main text).** The ET pathway is reported here with its caveats and carries no evidential weight in the paper’s conclusions. Whole-basin ET fails its event-study pre-trend test (differential pre-2010 trend  $p$  approximately  $3e-5$ ) and is confounded by residual aridity over barren land; masking barren pixels removes both the pre-trend failure and the positive effect (Supplementary Fig. S7), but the resulting orchard stratum was defined post hoc and relies on MOD16, whose documented underestimation of ET over irrigated land biases its slope gap toward zero, disqualifying a null as evidence in a null-finding design. A thermal ET product (Landsat SSEBop/METRIC or ECOSTRESS class) is the way to test the consumptive-use pathway properly and is flagged as future work.

| Group                              | Outcome                | Estimand                           | Estimate | 95% CI             | p    |
|------------------------------------|------------------------|------------------------------------|----------|--------------------|------|
| Cross-sectional matched ATT        | Reservoir ET buffering | d log ET / d SPEI                  | -0.014   | [-0.0646, 0.0369]  | -    |
| Forcing-interacted DiD (slope-gap) | Whole-basin ET         | differential deficit->impact slope | 0.048    | [0.0255, 0.0703]   | 0.11 |
| Forcing-interacted DiD (slope-gap) | Orchard ET             | differential deficit->impact slope | -0.020   | [-0.0448, 0.00565] | 0.24 |

**Supplementary Table S17 | Groundwater-substitution bound from the DGA well hydrograph network.** Differential groundwater drawdown between dammed and matched-control basins over the megadrought (2010–2021). Each well’s depletion rate is the ordinary least squares slope of Hampel-filtered depth to water on time ( $\text{m yr}^{-1}$ , positive = falling water table); per-basin rates are aggregated by the outlier-robust median (primary) and, for robustness, the mean. **dammed** / **control** are the weighted group-mean rates; **att\_wo** is the entropy-balancing weighting-only ATT (with 95% CI and stratified randomization-inference **perm\_p**) and **att\_dr** the aridity-adjusted doubly-robust ATT. Coverage is 26 matched basins (13 dammed, 13 control, 213 wells). The differential is not significant in either specification, and the depletion residuals carry no spatial autocorrelation (**moran\_I**, **moran\_p**), so dammed basins show no faster groundwater drawdown of the kind a substitution confound would require. One scope limitation applies: the DGA monitoring network is not designed around agriculture, so monitored wells need not sit inside the high-value orchard footprints where compensatory pumping concentrates, nor sample the same aquifer strata. The basin-median rates therefore bound basin-scale substitution; field-scale pumping localized outside the monitored network remains a residual caveat.

| Spec                 | Dammed<br>(m/yr) | Control<br>(m/yr) | ATT<br>(WO) | 95% CI           | p_perm | ATT<br>(DR) | Moran<br>I | Moran<br>p | wells |
|----------------------|------------------|-------------------|-------------|------------------|--------|-------------|------------|------------|-------|
| median<br>(primary)  | 0.098            | 0.079             | 0.089       | [-0.17,<br>0.35] | 0.681  | 0.028       | -<br>0.069 | 0.752      | 213   |
| mean<br>(robustness) | 0.289            | 0.076             | 0.283       | [-0.11,<br>0.67] | 0.328  | 0.486       | -<br>0.014 | 0.726      | 213   |

**Supplementary Table S18 | Pre-drought aridity-window sensitivity of the matching.**

The 1991–2020 World Meteorological Organization normal used for baseline aridity overlaps eleven megadrought years, so the covariate could in principle absorb post-treatment variation. Recomputing unit aridity on the strictly pre-drought 1991–2009 window leaves the design unchanged: unit aridity is essentially identical across windows (Pearson and Spearman rows), 99% of matched units keep the aridity tercile used as a permutation stratum, and the megadrought change in P/PET (2010–2020 minus 1991–2009) does not differ significantly between treated and weighted control basins and is an order of magnitude smaller than the treated-control aridity gap the matching addresses. The matching therefore does not lean on drought-period or reservoir-influenced variation.

| Quantity                                                                       | Value    | Detail                                                                                  |
|--------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------|
| unit aridity, pre-drought (1991-2009) vs full (1991-2020) window: Pearson r    | 0.9982   | Spearman rho 0.9993 (n=265 matched units)                                               |
| aridity-tercile agreement between windows (permutation strata)                 | 0.9920   | share of matched units assigned the same tercile                                        |
| median P/PET, treated: pre-drought vs full window                              | 0.2730   | full 0.227; control pre 0.798, full 0.680                                               |
| megadrought change in P/PET (2010-2020 minus 1991-2009), treated minus control | - 0.0441 | SE 0.0265, p = 0.10 (ebal-weighted); small against the ~0.5 treated-control aridity gap |

**Supplementary Table S19 | Aridity-overlap sensitivity.** Key results re-estimated on the sub-sample of basins whose baseline log-aridity lies in the treated/control overlap band (all 21 treated basins and 158 of 244 controls fall inside it), to show the parametric aridity adjustment is not masking a true effect where common support is genuine. The streamflow transmission-slope gap is essentially unchanged, and the water-rights expansion ATT shrinks and now straddles zero.

| Quantity                       | Full sample | Aridity overlap | Overlap 95% CI |
|--------------------------------|-------------|-----------------|----------------|
| Streamflow SPEI->SSI slope gap | -0.183      | -0.174          | -              |
| Water-rights expansion ATT     | 20.036      | 11.880          | [-5.62, 29.4]  |



**Supplementary Table S20 | Fully non-parametric matching on the aridity-overlap subset.** Every cross-sectional outcome re-estimated after re-fitting entropy balancing *within* the treated/control aridity-overlap band and using the weighting-only estimator (ebal-weighted difference in means, no outcome-model aridity term), so the estimate performs no parametric extrapolation across the aridity regime gap. Cropland-area expansion, orchard-ET buffering and the water-rights count remain null. The winsorized water-rights volume is the only outcome whose stratified-permutation p falls below 0.05 here; Supplementary Table S27 shows this reflects spatial autocorrelation, not a reservoir effect. The **perm p** column is the unit-level stratified randomization-inference p (Köppen  $\times$  aridity tercile).

| Outcome                             | ATT     | 95% CI (HC3)       | p_perm | n_t | n_c |
|-------------------------------------|---------|--------------------|--------|-----|-----|
| Cropland-area expansion (ha/km2)    | -0.5860 | [-2.08, 0.905]     | 0.328  | 21  | 158 |
| Orchard ET buffering (dlogET/dSPEI) | -0.0182 | [-0.0427, 0.00631] | 0.291  | 18  | 30  |
| Water-rights count (per 100 km2)    | 12.0000 | [-4.36, 28.4]      | 0.420  | 21  | 136 |
| Water-rights volume (l/s per km2)   | 1.8700  | [0.164, 3.58]      | 0.005  | 21  | 136 |

**Supplementary Table S21 | Metric and functional-form checks.** Top: the baseline SPEI-to-SSI transmission slope among undammed controls varies smoothly and monotonically across aridity terciles, with no regime break for the linear log-aridity adjustment to misfit. Middle: the control elevation sensitivity of transmission shows no detectable nonlinearity across the ~2000 m snowline (piecewise interaction p = 0.25); the snow regime itself is tested directly on the placebo estimand in Supplementary Table S15. Bottom: the cropland null is metric-invariant: re-fit on log cropland share (relative rather than absolute expansion), the forcing-interacted difference-in-differences slope gap remains permutation-null.

| Quantity                                                                           | Value    | Detail                                                                            |
|------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------|
| control baseline SPEI->SSI slope by aridity tercile (arid / mid / humid)           | 0.6470   | mid 0.699, humid 0.743; smooth monotone variation, no regime break                |
| control elevation sensitivity of transmission, below vs above the ~2000 m snowline | - 0.0780 | above 0.212 per km; interaction p = 0.25 (n=84 control gauges)                    |
| relative cropland expansion: forcing-DiD slope gap on log cropland share           | 0.0473   | SE 0.046, perm p = 0.369 (140 units with nonzero cropland); metric-invariant null |

**Supplementary Table S22 | Non-linear aridity-by-forcing sensitivity.** Drought transmission need not be linear in aridity, so the residual-aridity adjustment is not left to the linear outcome model: the buffering model adds quadratic aridity-by-forcing interactions. The curvature is strongly detectable (quadratic term  $p < 10^{-5}$ ), and allowing it collapses the apparent intent-to-treat buffering from  $-0.18$  to  $-0.06$  (permutation  $p = 0.43$ ), a third independent route, alongside the within-region estimator and the hard balance, by which flexible aridity adjustment absorbs the apparent effect. This is the signature of a non-linear siting confound, not of regulation.

| Quantity                                                | ValueDetail                                                                                                                                                                |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| buffering ITT net of quadratic aridity-by-forcing terms | - perm p = 0.430 (1000 perms); linear-adjustment primary -0.183<br>0.057                                                                                                   |
| aridity-by-forcing terms (linear; quadratic)            | 0.033linear p = 0.364; quadratic -0.067, p = 1.8e-06 (cluster-robust); the control transmission gradient across aridity is also tercile-checked in Supplementary Table S21 |

**Supplementary Table S23 | Hard-balanced aridity sensitivity.** The primary design deliberately leaves a residual aridity imbalance (SMD 0.17) to the doubly-robust outcome model because hard-balancing aridity collapses the effective control sample from 91 to about 19. Here the strict alternative is reported: entropy balancing that targets log baseline aridity alongside log area and elevation within Köppen group, achieving SMD 0.000 with no parametric extrapolation across the hydroclimatic regime gap. Every headline result is unchanged or strengthened: the streamflow slope gap and the upstream placebo collapse essentially to zero, and the cropland and water-rights outcomes remain null. That even the *apparent* buffering vanishes when aridity is fully balanced is the direct signature of a siting confound rather than a reservoir effect. The weight concentration this balance requires, and a leave-one-out sensitivity showing the point-estimate vanishing depends on the single most-weighted control basin, are disclosed in Supplementary Table S24; the hard balance is corroborating, not load-bearing.

| Quantity                                                              | Value      | Detail                                                                      |
|-----------------------------------------------------------------------|------------|-----------------------------------------------------------------------------|
| effective control sample size (aridity hard-balanced)                 | 18.900000  | log-aridity SMD after balancing 0.000 (residual 0.17 in the primary design) |
| streamflow treat x SPEI, ITT                                          | - 0.023000 | SE 0.160, perm p = 0.930 (1000 perms); primary design -0.183                |
| streamflow treat x SPEI, upstream placebo                             | - 0.047000 | SE 0.154, perm p = 0.882; primary design -0.20                              |
| cropland-area DiD slope gap                                           | 0.000508   | SE 0.0023, perm p = 0.855; primary design perm p = 0.91                     |
| water-rights expansion ATT (weighting-only, per 100 km <sup>2</sup> ) | - 4.900000 | 95% CI [-36.9, 27.1], perm p = 0.855                                        |

**Supplementary Table S24 | Hard-balance weight diagnostics and leave-one-out sensitivity.** The aridity hard balance achieves SMD 0.000 at the cost of concentrated control weights (maximum normalized share 13%, top five 44%, Kish effective sample 19 of 244), so its stability is tested two ways. The naive leave-one-out drops a top-weighted control with the remaining weights unchanged; because the dropped unit carried part of the aridity balance, the estimate mechanically drifts back toward the confounded primary value, which is expected and is not evidence of fragility. The design-consistent re-balanced leave-one-out re-fits the entropy balance without each unit (log-aridity SMD stays 0.000): the estimate stays near zero for seven of the eight top-weighted controls, but dropping the single most-weighted basin returns the confounded value with balance restored, so the hard-balanced point estimate depends on one control basin. No variant turns the null into detected buffering, since every value in the span lies inside the primary design's permutation null; the hard balance is accordingly read as a fragile corroboration, with the causal weight on the within-region comparison and the upstream placebo, which use no aridity balance.

| Quantity            | ValueDetail                                                                      |
|---------------------|----------------------------------------------------------------------------------|
| hard-balance        | 0.132top-5 share 0.44, Kish ESS 18.9 of 244 controls (participation ratio 0.077) |
| control weights:    |                                                                                  |
| maximum             |                                                                                  |
| normalized share    |                                                                                  |
| hard-balanced       | - primary-design estimate -0.183; hard-balanced perm p > 0.85                    |
| ITT slope, all      | 0.023(Supplementary Table S23)                                                   |
| controls            |                                                                                  |
| naive               | 0.182estimates span [-0.182, 0.057]; largest excursion dropping unit 0250;       |
| leave-one-out       | removing a balancing unit without re-balancing reintroduces the aridity          |
| (weights fixed)     | imbalance, so drift toward the confounded -0.183 is mechanical, not              |
| over top-8 controls | fragility                                                                        |
| re-balanced         | 0.184entropy balance re-fit without each unit: estimates span [-0.184, 0.002],   |
| leave-one-out over  | log-aridity SMD stays -0.000 to 0.000, ESS 11.6 to 18.1; largest excursion       |
| top-8 controls      | dropping unit 0250                                                               |

**Supplementary Table S25 | Buffering under a smooth national time trend.** Year fixed effects absorb the common temporal component of the megadrought, so the primary slope gap is identified from within-year spatial variation in SPEI. Replacing the year fixed effects with a cubic national time trend (month effects retained) restores the temporal drought signal to the identifying variation; every slope is reproduced and remains permutation-null, so the null is not an artifact of the fixed-effects structure or of limited within-year variation.

| Quantity                                  | Value   | Detail                                                                                                                 |
|-------------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------|
| smooth-trend buffering, ITT               | - 0.188 | perm p = 0.320 (1000 perms); year-FE primary -0.183                                                                    |
| smooth-trend buffering, downstream        | - 0.170 | perm p = 0.445 (1000 perms); year-FE primary -0.165                                                                    |
| smooth-trend buffering, upstream placebo  | - 0.190 | perm p = 0.487 (1000 perms); year-FE primary -0.201                                                                    |
| smooth-trend placebo contrast (down - up) | 0.021   | cubic national time trend + month FE retain the temporal drought signal that year FE absorb; regulation would give < 0 |

**Supplementary Table S26 | Physical-volume translation of the equivalence margin.**

The  $\pm 25\%$  equivalence margin corresponds at the worst observed deficit to a perturbation of 0.29 SSI units (Methods). Translated through each treated downstream gauge's within-gauge linear sensitivity of the 12-month rolling mean flow to its SSI, that perturbation corresponds to the volumes below, expressed per 12-month season and as a share of mean annual flow. The linear sensitivity is an average rather than a dry-tail mapping and therefore overstates the worst-drought volume; even so, the median is of order tens of cubic hectometres, a volume that is material for local water management. This is why the margin is anchored to the operational drought-severity classification rather than to volumetric negligibility, why the threshold-free minimum detectable effect is reported alongside, and why the causal weight of the null rests on the within-basin placebo rather than on the equivalence verdict.

| Quantity                                                                       | Value | Detail                                                                                                                                          |
|--------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| equivalence-margin volume, treated downstream gauges (hm3 per 12-month season) | 58.0  | median across 40 gauges; IQR [16.3, 137.2]; flow-SSI sensitivity x 0.29 SSI x 31.5; linear (average) sensitivity overstates the dry-tail volume |
| equivalence-margin volume as % of mean annual flow                             | 10.7  | IQR [6.9, 14.0]%                                                                                                                                |

**Supplementary Table S27 | Spatial autocorrelation and spatially-restricted inference.**

Top: Moran's I of each outcome's ATT-model residuals (great-circle distances, five-nearest-neighbour row-standardized weights, 999 residual permutations). The water-rights outcomes and cropland area show significant positive spatial autocorrelation, which the unit-level stratified permutation does not preserve. Bottom: for the water-rights outcomes, the ATT with its spatially-naive HC3 heteroskedasticity-consistent interval and stratified-permutation p alongside the cuenca-block-clustered interval and the spatial-block permutation p (treatment swapped at the cuenca level). The apparent water-rights volume effect on the overlap subset (stratified p = 0.005) is not significant once spatial dependence is respected (spatial-block p = 0.077; cuenca-clustered interval spans zero), and the count is null under both.

| Outcome                 | n   | Moran's I | Expected | p     |
|-------------------------|-----|-----------|----------|-------|
| Cropland-area expansion | 265 | 0.181     | -0.004   | 0.001 |
| Water-rights count      | 216 | 0.599     | -0.005   | 0.001 |
| Water-rights volume     | 216 | 0.388     | -0.005   | 0.001 |
| Orchard ET buffering    | 49  | 0.069     | -0.021   | 0.198 |

| Outcome                              | ATT   | 95% CI<br>(HC3)   | p<br>(stratified) | 95% CI<br>(cuenca) | p (spatial<br>block) |
|--------------------------------------|-------|-------------------|-------------------|--------------------|----------------------|
| Water-rights volume (l/s<br>per km2) | 1.87  | [0.16, 3.58]      | 0.005             | [-0.33, 4.07]      | 0.077                |
| Water-rights count (per 100<br>km2)  | 12.00 | [-4.36,<br>28.36] | 0.420             | [-2.70, 26.70]     | 0.581                |

**Supplementary Table S28 | Within-region residual (siting-ladder rung 4), permutation inference.** The design estimator with the baseline SPEI-to-outcome slope allowed to vary by Köppen region, the “regulation estimate” left after the between-region aridity gradient is removed (rung 4 of Supplementary Fig. S6 and Table S47), reported with its own permutation null: the residual is  $-0.05$  ( $p = 0.65$ ).

| Quantity                                                     | Esti-<br>mate | Detail                                                        |
|--------------------------------------------------------------|---------------|---------------------------------------------------------------|
| siting-ladder rung 4 (design + within region), permutation p | -0.05         | est -0.050, permutation p = 0.652 (2000 perms), null SD 0.110 |

**Supplementary Table S29 | Direct test of the upstream/downstream placebo contrast.** The paper’s central falsifier, tested directly rather than asserted from the separate point estimates:  $D = \beta_{\text{down}} - \beta_{\text{up}}$ , the downstream-minus-upstream difference in the differential (treat  $\times$  SPEI) transmission slope, against its permutation null (treatment reassigned at the unit level within Köppen  $\times$  aridity-tercile strata, the same draw applied to both gauge-class panels; Methods). A genuine regulation effect predicts  $D < 0$ ; the observed contrast is  $+0.037$  ( $p = 0.72$ ), so the regulated reaches show no attenuation beyond the unregulated placebo of the same basins.

| Quantity                                                                 | Es-<br>ti-<br>mate | Detail                                                                                                                                                       |
|--------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| within-basin placebo contrast (downstream - upstream differential slope) | 0.037              | downstream -0.165, upstream placebo -0.201, $D = +0.037$ (SE_perm 0.098); permutation p = 0.718 (2000 perms); a genuine regulation effect would give $D < 0$ |

**Supplementary Table S30 | Elevation-partitioned permutation null for the placebo contrast.** In the composition-fixed permutation scheme a pseudo-treated control basin contributes an identical gauge set to both panels, so natural up/down heterogeneity is absent from its null contribution, which could compress the null variance of  $D$  and make the test anti-conservative (a bias that can only overstate significance, and therefore cannot manufacture the paper’s null). Re-testing under a variant null in which each pseudo-treated control basin’s gauges are partitioned at its median gauge elevation shows no compression: the aggregate null SD is unchanged (0.093 against 0.098) with  $p = 0.73$ , and the carryover contrast survives the corrected null ( $p = 0.025$  against 0.014).

| Quantity                                              | Value   | Detail                                                                                                                   |
|-------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------|
| aggregate contrast D under elevation-partitioned null | 0.037   | perm p = 0.728, null SD 0.093 (1000 perms); composition-fixed scheme: p = 0.718, null SD 0.098 (Supplementary Table S29) |
| carryover contrast D under elevation-partitioned null | - 0.232 | perm p = 0.025, null SD 0.103 (1000 perms); composition-fixed scheme: p = 0.014 (Supplementary Table S50)                |

**Supplementary Table S31 | Upstream/downstream placebo comparability.** Upstream gauges are higher than downstream, and gauge elevation predicts the transmission slope among undammed controls ( $-0.21$  per  $+1000$  m), yet upstream and downstream treated gauges have nearly identical raw transmission slopes, so the equal up/down attenuation is not an elevation artifact.

| quantity                                                | value    | detail                                                                                                                                        |
|---------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| upstream gauges (elev, mean m)                          | 1003.000 | n=28; mean transmission 0.565                                                                                                                 |
| downstream gauges (elev, mean m)                        | 459.000  | n=40; mean transmission 0.599                                                                                                                 |
| control transmission slope vs elevation (per $+1000$ m) | - 0.212  | p = 0.00 (n=84 control gauges); elevation predicts transmission, so the up/down gap is bounded with this sensitivity rather than assumed away |

**Supplementary Table S32 | Catchment-area comparability of the placebo.** The unit-area gap between upstream and downstream gauges is negligible, and log unit area does not predict the transmission slope among undammed control gauges (with a tercile check for non-monotonicity), so catchment size cannot manufacture the upstream attenuation. The unit area is the sub-watershed polygon each gauge sits in; the cumulative-scale confound, which the polygon area does not capture, is tested with gauge discharge in Supplementary Table S34.

| Quantity                                            | Value   | Detail                                                                                                                                           |
|-----------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| upstream gauges (unit area, km <sup>2</sup> )       | 3160.00 | n=28; mean transmission 0.565                                                                                                                    |
| downstream gauges (unit area, km <sup>2</sup> )     | 3070.00 | n=40; mean transmission 0.599                                                                                                                    |
| control transmission slope vs log(area), per e-fold | - 0.07  | p = 0.06 (n=84 control gauges); area does not predict transmission, so the up/down catchment-size gap cannot manufacture the placebo attenuation |
| control transmission by area tercile (mean slope)   | NA      | small 0.865 (n=28)   mid 0.891 (n=28)   large 0.725 (n=28)                                                                                       |

**Supplementary Table S33 | Elevation-adjusted placebo.** The buffering model re-fit with a SPEI-by-elevation interaction, so the differential slope is net of the elevation-transmission gradient (which alone would make upstream reaches look *more* buffered; main text). The elevation term itself is strongly detectable (`elev_term_p`), yet every treatment effect and the down-minus-up contrast remain null, so the placebo result is not an elevation artifact.

| Subset           | Raw est.   | Adjusted est. | p_perm (adj) | Elev. term p | n perms |
|------------------|------------|---------------|--------------|--------------|---------|
| ITT              | -0.1828730 | -0.1952412    | 0.2998501    | 0.00043      | 2000    |
| downstream       | -0.1645997 | -0.1960080    | 0.3308346    | 0.00074      | 2000    |
| upstream placebo | -0.2014380 | -0.2117479    | 0.3903048    | 0.00088      | 2000    |



| Subset           | Raw est. | Adjusted est. | p_perm (adj) | Elev. term p | n perms |
|------------------|----------|---------------|--------------|--------------|---------|
| contrast down-up | NA       | 0.0157399     | 0.8905547    | -            | 2000    |

**Supplementary Table S34 | Cumulative-scale (discharge-adjusted) placebo.** Because larger cumulative catchment scale genuinely flattens transmission among controls (+0.08 per log discharge,  $p < 0.001$ ), the buffering model adds a SPEI-by-log-discharge interaction so the differential slope is net of cumulative scale. Every effect and the contrast remain null, and the scale gap is small in most paired basins (median downstream/upstream discharge ratio 1.1), because downstream gauges sit just below the dams.

| Quantity                                                       | ValueDetail                                                                                                                                               |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| scale-adjusted buffering, ITT (SPEI x log discharge)           | - raw -0.183; permutation $p = 0.502$ (1000 perms)<br>0.093                                                                                               |
| scale-adjusted buffering, downstream                           | - raw -0.165; permutation $p = 0.504$ (1000 perms)<br>0.102                                                                                               |
| scale-adjusted buffering, upstream placebo                     | - raw -0.201; permutation $p = 0.678$ (1000 perms)<br>0.072                                                                                               |
| scale-adjusted placebo contrast (down - up)                    | - permutation $p = 0.737$ (1000 perms); regulation would give $< 0$<br>0.030                                                                              |
| control transmission slope vs log discharge (cumulative scale) | 0.078 $p = 0.000$ ( $n=84$ control gauges); larger cumulative scale flattens transmission, so the contrast must be scale-adjusted                         |
| downstream/upstream mean-discharge ratio (paired basins)       | 1.110 median 1.11, IQR [0.94, 2.26], $n=15$ ; 10 of 15 basins have ratio $< 2$ , so downstream cumulative area is NOT vastly larger in most paired basins |

**Supplementary Table S35 | Geomorphological (alluvial) confound check on the placebo.**

The up/down contrast coincides with the bedrock-to-alluvial transition, and elevation is only a proxy for it, so a direct terrain metric is extracted: local relief, the maximum minus minimum Shuttle Radar Topography Mission (SRTM) elevation within 2.5 km of each gauge, a standard discriminator of bedrock canyons (high relief) from alluvial valleys (low relief). The up/down relief gap is real (about 0.7 SD), but relief does not predict the transmission slope among undammed controls ( $p = 0.41$ ; marginal at  $p = 0.08$  net of elevation), a relief-by-forcing adjustment leaves both placebo coefficients essentially unchanged, and among low-relief (alluvial) units neither the upstream nor the downstream contrast shows any buffering (both positive and null, on few treated units). Alluvial storage therefore neither manufactures the upstream attenuation nor masks a downstream reservoir effect; relief at 2.5 km cannot resolve aquifer storage volumes themselves, which remain a residual caveat.

| Quantity                                                          | Value       | Detail                                                                    |
|-------------------------------------------------------------------|-------------|---------------------------------------------------------------------------|
| gauges: up vs down local relief (m, 2.5 km radius)                | 796.0000    | down 502 (gap 294 m = 0.72 SD); n_up=28, n_down=32                        |
| control gauges: transmission slope per SD relief                  | 0.0263      | SE 0.0320, $p = 0.413$ (n=84); net of elevation $p = 0.084$               |
| upstream placebo: treat x SPEI, unadjusted                        | -<br>0.2014 | SE 0.0904, $p = 0.030$                                                    |
| upstream placebo: treat x SPEI, relief-adjusted (spei x relief z) | -<br>0.1968 | SE 0.0912, $p = 0.035$ ; relief term -0.0132 (SE 0.0403, $p = 0.745$ )    |
| upstream placebo: treat x SPEI, low-relief (alluvial) units       | 0.0602      | SE 0.1664, $p = 0.721$ ; 3 treated / 26 control units below median relief |
| downstream: treat x SPEI, unadjusted                              | -<br>0.1646 | SE 0.0851, $p = 0.058$                                                    |
| downstream: treat x SPEI, relief-adjusted (spei x relief z)       | -<br>0.2035 | SE 0.1052, $p = 0.058$ ; relief term -0.0483 (SE 0.0439, $p = 0.276$ )    |
| downstream: treat x SPEI, low-relief (alluvial) units             | 0.1250      | SE 0.1268, $p = 0.333$ ; 7 treated / 22 control units below median relief |

**Supplementary Table S36 | Gauge-classification robustness of the placebo contrast.**

The contrast re-estimated (i) excluding downstream gauges whose DGA hydrographic sub-cuenca is named for a different river or an unregulated tributary than the dam and (ii) excluding downstream gauges within 100 m of the dam elevation, the ones most plausibly on a parallel tributary. The contrast stays null in every scenario, and the downstream estimate moves toward zero rather than toward buffering.

| Scenario                            | n<br>down | Down        | Up        | D         | p_D   | p<br>(down) |
|-------------------------------------|-----------|-------------|-----------|-----------|-------|-------------|
| baseline (all downstream gauges)    | 61        | - 0.1645997 | -0.201438 | 0.0368383 | 0.723 | 0.481       |
| excluding cross-river gauge(s)      | 59        | - 0.1606022 | -0.201438 | 0.0408358 | 0.688 | 0.494       |
| excluding gauges within 100m of dam | 45        | - 0.0823384 | -0.201438 | 0.1190996 | 0.470 | 0.504       |

**Supplementary Table S37 | Routed-network validation of the upstream/downstream classification.**

The gauge classification is by SRTM elevation, so “upstream” gauges could in principle sit on parallel, hydrologically disconnected tributaries. Every treated gauge and every dam is snapped to its nearest HydroRIVERS v1.0 reach (all 96 gauges snap, median 264 m; all 26 dams, median 124 m) and the NEXT\_DOWN topology traced: an upstream gauge is network-connected if its flow path passes its dam’s reach, a downstream gauge if it lies on the dam’s routed downstream path. HydroRIVERS carries a finite drainage threshold, so a gauge that fails the trace mixes true parallel placement with reaches too small for the network; the connected subset is therefore the clean test rather than the unconnected set proof of disconnection. On strictly network-validated gauges the placebo verdict strengthens: the connected-only upstream slope deepens to  $-0.36$ , and the fully validated same-channel contrast is *positive* (down  $-0.24$ , up  $-0.36$ ,  $D = +0.12$ ,  $p = 0.18$ ), the unregulated reaches more attenuated than the regulated ones, the opposite sign of a regulation effect.

| Quantity                                                  | Value Detail                                                                                                                           |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| gauges snapped to HydroRIVERS reaches                     | 96.000 of 96 treated gauges with coordinates; median snap distance 264 m; 26 dams snapped (median 124 m); snaps beyond 5000 m excluded |
| upstream gauges network-connected through their dam       | 15.000 of 35 upstream gauges: flow path (NEXT_DOWN trace) passes the dam reach; the remainder are parallel tributaries or unsnapped    |
| downstream gauges on the dam’s routed downstream path     | 16.000 of 61 downstream gauges                                                                                                         |
| upstream placebo slope, network-connected gauges only     | - perm $p = 0.265$ ; contrast vs full downstream panel $D = 0.360 + 0.196$ , $p = 0.116$ (1000 perms); raw upstream $-0.20$            |
| fully network-validated contrast (both classes connected) | 0.123 down $-0.238$ , up $-0.360$ , perm $p = 0.182$ (1000 perms)                                                                      |

### Supplementary Table S38 | Cascade-dam contamination screen of the upstream placebo.

The placebo assumes upstream reaches are physically unregulated, which cascade systems can violate (Ralco above Pangué on the Biobío; the Laguna del Maule works above Colbún; La Laguna above Puclaro on the Elqui): an upstream gauge below another dam would carry real regulation misread as siting. Every upstream gauge is screened against the 1,370-dam national inventory. A *material* dam is DGA-monitored, registry-classified GRANDE, or of wall height at least 15 m (the International Commission on Large Dams threshold); dam elevations are SRTM, the paper's classification convention. The within-sub-watershed tier flags gauges below a material dam inside their own treated sub-watershed; the named-cascade tier adds the gauges below the historically documented upstream works on the same river systems (La Laguna 1937 above Puclaro; the Laguna del Maule works 1951 above Colbún; Lagunas del Huasco 1911 above Santa Juana), the true cascades that sit outside the treated sub-watershed; and the same-cuenca tier flags gauges below any material dam anywhere in their DGA cuenca, a conservative bound that necessarily over-flags parallel tributaries. The placebo survives all three exclusions, and the direction of the change is diagnostic: hidden upstream regulation predicts the cleaned upstream slope should steepen toward zero attenuation, and instead it deepens slightly. Under the strict tier-2 exclusion the carryover contrast keeps its direction and grows while its permutation p rises on the thinned upstream panel, consistent with its exploratory label.

| Quantity                                                               | Value  | Detail                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| upstream gauges screened                                               | 35.000 | cross 16 treated basins; screened against 445 inventory dams inside treated basins and 54 material dams (monitored / GRANDE / wall $\geq 15$ m) in the same cuencas, from the 1,370-dam national inventory                                                                                                                        |
| tier 1: below a material dam in the same sub-watershed                 | 3.000  | units: 0733, 0831                                                                                                                                                                                                                                                                                                                 |
| named-cascade tier: tier 1 plus gauges below documented upstream works | 10.000 | adds gauges below La Laguna (1937, above Puclaro), the Laguna del Maule works (1951, above Colbun), and Lagunas del Huasco (1911, above Santa Juana); units: 0382, 0432, 0732, 0733, 0831                                                                                                                                         |
| tier 2: below any material dam in the same cuenca (conservative bound) | 20.000 | units: 0382, 0432, 0450, 0471, 0542, 0573, 0603, 0732, 0733, 0831; units losing all upstream gauges: 0382, 0432, 0471, 0542, 0573, 0603, 0732, 0733, 0831; includes true cascades (Laguna del Maule 1951 above Colbún; La Laguna 1937 above Puclaro; Lagunas del Huasco 1911 above Santa Juana) and parallel-tributary over-flags |
| upstream placebo slope, tier-1-cleaned                                 | 0.245  | - perm p = 0.280; contrast D = +0.081, p = 0.448 (1000 perms); raw upstream -0.20, raw D = +0.04                                                                                                                                                                                                                                  |
| upstream placebo slope, named-cascade-cleaned                          | 0.204  | - perm p = 0.357; contrast D = +0.039, p = 0.792 (1000 perms); n upstream gauges retained 25 of 35                                                                                                                                                                                                                                |
| upstream placebo slope, tier-2-cleaned                                 | 0.213  | - perm p = 0.471; contrast D = +0.048, p = 0.817 (1000 perms); n upstream gauges retained 15 of 35                                                                                                                                                                                                                                |

| Quantity                                               | Value  | Detail                                                                                 |
|--------------------------------------------------------|--------|----------------------------------------------------------------------------------------|
| carryover<br>contrast D,<br>tier-2-cleaned<br>upstream | -0.359 | - down -0.459, up -0.100, perm p = 0.127 (n_treat = 7); baseline D = -0.232, p = 0.014 |

**Supplementary Table S39 | Non-linear transmission test.** A quadratic SPEI term interacted with treatment on the full panel: a buffering effect confined to extreme deficits, as threshold-based release rules would produce, would appear as a nonzero  $\text{treat} \times \text{SPEI}^2$  coefficient. It is null.

| Quantity                                                                                  | Es-<br>ti-<br>mate | Detail                                                                                                                                      |
|-------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| non-linear transmission test<br>( $\text{treat} \times \text{SPEI}^2$ ),<br>permutation p | -<br>0.025         | est -0.025, permutation p = 0.654 (2000 perms), null SD<br>0.042; a dry-tail-only buffering effect would appear as a<br>nonzero interaction |

**Supplementary Table S40 | Dry-tail re-estimates of the buffering and placebo tests.** The transmission models and the up/down contrast re-fit on meteorological drought months only ( $\text{SPEI} < -1.0$ , moderate drought and drier; and  $\text{SPEI} < -1.5$  as a lower-power sensitivity), each under permutation inference. No subset shows a downstream-specific effect beyond its upstream placebo; the  $< -1.5$  rows retain few months per gauge and are reported for completeness.

| Quantity                                                            | Value      | Detail                                                                           |
|---------------------------------------------------------------------|------------|----------------------------------------------------------------------------------|
| dry-tail buffering, ITT ( $\text{SPEI} < -1.0$ )                    | 0.001      | est +0.001, permutation p = 1.000 (2000 perms)                                   |
| dry-tail buffering, downstream ( $\text{SPEI} < -1.0$ )             | -<br>0.199 | est -0.199, permutation p = 0.711 (2000 perms)                                   |
| dry-tail buffering, upstream placebo<br>( $\text{SPEI} < -1.0$ )    | 0.052      | est +0.052, permutation p = 0.939 (2000 perms)                                   |
| dry-tail placebo contrast<br>down-minus-up ( $\text{SPEI} < -1.0$ ) | -<br>0.250 | D = -0.250, permutation p = 0.487 (2000 perms);<br>regulation would give $D < 0$ |
| dry-tail buffering, ITT ( $\text{SPEI} < -1.5$ )                    | 0.584      | est +0.584, permutation p = 0.512 (2000 perms)                                   |
| dry-tail buffering, downstream ( $\text{SPEI} < -1.5$ )             | 2.518      | est +2.518, permutation p = 0.425 (2000 perms)                                   |
| dry-tail buffering, upstream placebo<br>( $\text{SPEI} < -1.5$ )    | -<br>0.709 | est -0.709, permutation p = 0.518 (2000 perms)                                   |
| dry-tail placebo contrast<br>down-minus-up ( $\text{SPEI} < -1.5$ ) | 3.227      | D = +3.227, permutation p = 0.453 (2000 perms);<br>regulation would give $D < 0$ |

**Supplementary Table S41 | Early/late megadrought phase split of the streamflow buffering and placebo tests.** A single 2005–2024 transmission slope could in principle average away an early-phase buffering signal (storage intact at drought onset) that fails later (storage depleted). Splitting the megadrought at 2015 shows the opposite of a depletion signature. In the early phase (2010–2014), when reservoirs entered near-full (peak 87% of capacity, Supplementary Table S52) and depletion cannot yet explain anything, the downstream and upstream placebo gaps are identical (treat x SPEI =  $-0.43$  in both), so the strong early attenuation is siting, exactly as in the pooled analysis. The attenuation then weakens from early to late phase, but this weakening is present, and is in fact sharpest, in the unregulated upstream placebo itself (shift  $+0.36$ ,  $p = 0.022$ ), which has no storage to deplete; it therefore reflects the changing forcing regime of the prolonged drought, not the exhaustion of reservoir buffering. At no phase does any subset show a downstream-specific effect beyond its upstream placebo. Per-phase p-values are cluster-robust and shown for completeness; the design’s primary inference remains the randomization test on the pooled estimand (main text).

| Quantity                                                          | Value       | Detail                                                                          |
|-------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------|
| ITT: treat x SPEI, early megadrought (2010-2014)                  | - 0.4442370 | SE 0.1235, $p = 0.001$ ; 17 treated / 44 control units                          |
| ITT: treat x SPEI, late megadrought (2015-2024)                   | - 0.3408339 | SE 0.2611, $p = 0.197$ ; 17 treated / 43 control units                          |
| ITT: early-to-late phase shift (treat x SPEI x late)              | 0.2879824   | SE 0.1544, $p = 0.067$ ; pooled 2010-2024 panel, year FE absorb the phase level |
| downstream: treat x SPEI, early megadrought (2010-2014)           | - 0.4299787 | SE 0.1028, $p = 0.000$ ; 15 treated / 44 control units                          |
| downstream: treat x SPEI, late megadrought (2015-2024)            | - 0.4441582 | SE 0.3616, $p = 0.224$ ; 14 treated / 43 control units                          |
| downstream: early-to-late phase shift (treat x SPEI x late)       | 0.2927161   | SE 0.1567, $p = 0.067$ ; pooled 2010-2024 panel, year FE absorb the phase level |
| upstream placebo: treat x SPEI, early megadrought (2010-2014)     | - 0.4301870 | SE 0.1589, $p = 0.009$ ; 15 treated / 44 control units                          |
| upstream placebo: treat x SPEI, late megadrought (2015-2024)      | - 0.2148096 | SE 0.2247, $p = 0.343$ ; 15 treated / 43 control units                          |
| upstream placebo: early-to-late phase shift (treat x SPEI x late) | 0.3643440   | SE 0.1547, $p = 0.022$ ; pooled 2010-2024 panel, year FE absorb the phase level |



**Supplementary Table S42 | Permutation inference for the megadrought phase shift.**

The early-to-late change of the buffering slope (treat  $\times$  SPEI  $\times$  late-phase interaction on the pooled megadrought panel; Methods) under the design's permutation inference, for the downstream panel, the upstream placebo, and their contrast. Neither phase shift is significant and the down-minus-up contrast is far from it; the largest shift is the upstream placebo's, which has no storage to deplete, so the phase evolution reflects the changing forcing regime of the prolonged drought rather than buffer exhaustion (companion to the per-phase subsample estimates in Supplementary Table S41).

| Quantity                                                                                          | Value Detail                                                                                             |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| downstream: early-to-late phase shift<br>(treat $\times$ SPEI $\times$ late), permutation p       | 0.293 est +0.293, permutation p = 0.355 (2000 perms)                                                     |
| upstream placebo: early-to-late phase<br>shift (treat $\times$ SPEI $\times$ late), permutation p | 0.364 est +0.364, permutation p = 0.088 (2000 perms)                                                     |
| phase-shift contrast (downstream shift -<br>upstream placebo shift)                               | - est -0.072, permutation p = 0.798 (2000 perms);<br>0.072 a regulation effect would give contrast $< 0$ |

**Supplementary Table S43 | Downstream extraction masking.** The up/down placebo compares regulated downstream reaches with unregulated upstream reaches of the same basins, and downstream valleys carry far more irrigation than upstream headwaters; drought-intensified downstream extraction could in principle steepen the downstream transmission slope and mask a real buffer. Rows: the downstream-versus-upstream local cropland gradient within the paired basins (MapBiomass class 18 in 5 km gauge buffers, aggregated per basin-class); the per-gauge control SPEI→SSI transmission slope regressed on local cropland fraction, with and without elevation and cumulative scale; and the masking magnitude predicted by the gradient times each sensitivity. The gradient is real (downstream gauges sit in roughly two and a half times the cropland of upstream gauges; 12 of 15 paired basins downstream-richer), but local extraction does not predict the transmission slope among control gauges, the point estimate carries the sign opposite to the masking mechanism, and even at face value the gradient predicts a slope shift near 0.01 SSI units, two orders of magnitude below the 0.28 contrast MDE.

| Quantity                                                               | Value       | Detail                                                                                                   |
|------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------|
| downstream-gauge local cropland (2007, 5 km)                           | 0.1974573   | per-basin mean over 15 paired basins; downstream valleys are ~2.5x more cropped than upstream headwaters |
| upstream-gauge local cropland (2007, 5 km)                             | 0.0774355   | per-basin mean over 15 paired basins                                                                     |
| down-up cropland gap (paired treated gauges)                           | 0.1200218   | positive = downstream more extracted; 12 of 15 paired basins have down > up                              |
| control transmission slope vs local cropland (raw)                     | - 0.0465798 | p = 0.749 (n=84 control gauges); local extraction does NOT predict transmission                          |
| control transmission slope vs local cropland (+ elevation)             | - 0.1002650 | p = 0.432; sign opposite to the masking mechanism                                                        |
| control transmission slope vs local cropland (+ elevation + discharge) | 0.0948221   | high-extraction reaches are if anything flatter                                                          |
| predicted masking magnitude (gap x raw sensitivity)                    | 0.0055906   | p = 0.407 (n=84); robust to scale/elevation                                                              |
| predicted masking magnitude (gap x full sensitivity)                   | 0.0113807   | = -0.0056 SSI/SPEI, far below the 0.28 MDE and the -0.23 carryover contrast                              |
|                                                                        |             | = -0.0114 SSI/SPEI, ~3% of the 0.28 MDE                                                                  |

**Supplementary Table S44 | Placebo net of a SPEI-by-local-cropland interaction.** The direct control for downstream extraction: the buffering model adds SPEI x local-cropland, so the differential slope is net of the extraction-driven transmission gradient, for the ITT, downstream, and upstream panels and the down-minus-up contrast, each under the design’s unit-level permutation inference. Adjusting for local extraction leaves the contrast null ( $D = -0.03$ ,  $p = 0.83$ ); the aggregate null is therefore not a masked buffer.

| Subset           | Estimate   | Perm. p   | n perms |
|------------------|------------|-----------|---------|
| ITT              | -0.2416300 | 0.1998002 | 1000    |
| downstream       | -0.2441015 | 0.2437562 | 1000    |
| upstream placebo | -0.2123475 | 0.5624376 | 1000    |
| contrast down-up | -0.0317540 | 0.8281718 | 1000    |

**Supplementary Table S45 | Sub-watershed SPEI exposure misclassification.** Upstream and downstream gauges within a paired basin sit in the same sub-watershed and receive the same subcuena-average SPEI-12, so each gauge’s true upslope-catchment exposure is misclassified to the extent that catchment differs from the subcuena mean. Rows: per gauge class, the correlation between gauge-point SPEI-12 (a conservative proxy for catchment exposure, since a catchment mean has lower error variance than any single cell) and the subcuena-mean SPEI the model uses, the error SD and variance, the SPEI variance, and the implied transmission-slope attenuation factor  $1/(1 + \text{var}(\text{err})/\text{var}(\text{SPEI}))$ . Exposure error is small and nearly symmetric (attenuation 0.95 downstream versus 0.93 upstream); the asymmetry biases the placebo contrast by at most about 0.01 units, and in the direction (upstream slightly more attenuated) that works against downstream- specific buffering.

| Class        | N     | r         | Error SD  | Error var | SPEI var  | Attenuation |
|--------------|-------|-----------|-----------|-----------|-----------|-------------|
| control      | 36300 | 0.9783510 | 0.1960651 | 0.0384415 | 0.8550878 | 0.9569779   |
| treated_down | 18300 | 0.9751562 | 0.2116526 | 0.0447968 | 0.8683456 | 0.9509421   |
| treated_up   | 10500 | 0.9671283 | 0.2418160 | 0.0584750 | 0.8298157 | 0.9341714   |

**Supplementary Table S46 | Paired-placebo contrast power.** The minimum detectable downstream-minus-upstream contrast at 80% power (two-sided  $\alpha = 0.05$ ), built from the permutation-null SD: 0.28 SSI units, about 52% of the downstream baseline transmission. The paired test therefore discriminates a siting signature from *large* regulation effects only; moderate downstream-specific buffering is not excluded, and the causal weight rests on the convergence of the elevation-, scale-, misclassification-, and extraction-adjusted contrasts rather than on any single test.

| Quantity                                                              | Value                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| paired-placebo contrast MDE (downstream minus upstream buffering gap) | 0.275 observed D + 0.037 (perm p 0.718), null SD 0.098; MDE 0.275 SSI units = 51.8% of the 0.53 downstream baseline at 80% power, alpha 0.05; a true downstream-specific buffering gap above this is detected, smaller gaps are not excluded (n = 15 paired basins) |

**Supplementary Table S47 | Siting-confound decomposition ladder (numeric).** The dammed-versus-control transmission-slope gap at each rung of Supplementary Figure S6, for streamflow SSI (intent-to-treat, ITT) and the two difference-in-differences proxy outcomes, plus the within-basin upstream placebo. `perm_p` is the design's randomization-inference p for the design rung and the placebo.

| Outcome                          | Rung                           | Slope gap | SE    | p_perm |
|----------------------------------|--------------------------------|-----------|-------|--------|
| Streamflow SSI (SPEI->SSI slope) | naive (no match, no FE)        | -0.197    | 0.090 | -      |
| Streamflow SSI (SPEI->SSI slope) | unit+time FE (unweighted)      | -0.165    | 0.075 | -      |
| Streamflow SSI (SPEI->SSI slope) | design (ebal + FE)             | -0.183    | 0.083 | 0.39   |
| Streamflow SSI (SPEI->SSI slope) | design + within region         | -0.050    | 0.070 | -      |
| Irrigated area (DiD)             | naive (no match, no FE)        | 0.001     | 0.003 | -      |
| Irrigated area (DiD)             | unit+time FE (unweighted)      | 0.002     | 0.002 | -      |
| Irrigated area (DiD)             | design (ebal + FE)             | 0.001     | 0.002 | 0.91   |
| Irrigated area (DiD)             | design + within region         | 0.001     | 0.002 | -      |
| Orchard ET (DiD)                 | naive (no match, no FE)        | -0.017    | 0.013 | -      |
| Orchard ET (DiD)                 | unit+time FE (unweighted)      | -0.015    | 0.013 | -      |
| Orchard ET (DiD)                 | design (ebal + FE)             | -0.020    | 0.013 | 0.24   |
| Orchard ET (DiD)                 | design + within region         | -0.020    | 0.012 | -      |
| Streamflow SSI (SPEI->SSI slope) | upstream placebo (unregulated) | -0.201    | 0.090 | 0.48   |

**Supplementary Table S48 | Positive control on the streamflow estimator.** A known buffering slope (`beta_injected`; negative = buffering) is added to treated downstream units; the estimator recovers it (recovered = baseline  $-0.17 + \text{beta}$ ), the no-effect case is not flagged, and randomization inference detects the effect once it exceeds the minimum detectable effect.

| beta_injected | recovered | p_perm | detected |
|---------------|-----------|--------|----------|
| 0.00          | -0.165    | 0.480  | FALSE    |
| -0.15         | -0.315    | 0.135  | FALSE    |
| -0.30         | -0.465    | 0.014  | TRUE     |
| -0.45         | -0.615    | 0.001  | TRUE     |

**Supplementary Table S49 | Equivalence / minimum-detectable-effect bounds on the headline nulls.** Observed slope gap, the minimum detectable effect (MDE; smallest effect detectable at 80% power from the permutation-null standard deviation, SD), the 90% (TOST) interval, and whether it lies within the negligible region ( $\pm 25\%$  of baseline). “baseline” is the untreated (control) SPEI→SSI / SPEI→impact transmission slope from the fixed-effects model. No outcome is formally equivalent to zero; the streamflow design rules out attenuations larger than  $\sim 46\text{--}59\%$  of baseline transmission.

| Outcome                     | Observed | MDE   | 90% CI              | Equiv. to 0? | p_perm |
|-----------------------------|----------|-------|---------------------|--------------|--------|
| streamflow SSI (ITT)        | -0.183   | 0.268 | [-0.34, -0.0253]    | no           | 0.39   |
| streamflow SSI (downstream) | -0.165   | 0.312 | [-0.348, 0.0186]    | no           | 0.48   |
| irrigated area (DiD)        | 0.001    | 0.010 | [-0.00506, 0.00656] | no           | 0.90   |
| orchard ET (DiD)            | -0.020   | 0.046 | [-0.0463, 0.00718]  | no           | 0.25   |

**Supplementary Table S50 | Storage-capacity (carryover) heterogeneity of the buffering test.** The buffering sample (17 treated basins with a usable downstream gauge) split by the reservoir carryover ratio (capacity / mean annual downstream flow, in years): ITT, downstream, and upstream slopes and the down-minus-up contrast  $D$  per subgroup, each under permutation inference. The seven carryover reservoirs (ratio  $\geq 0.5$ ) show a downstream-specific contrast ( $D = -0.23$ ,  $p = 0.014$ ) that the ten seasonal reservoirs do not ( $D = +0.18$ ,  $p = 0.26$ ). This split is an exploratory subgroup analysis and carries no multiplicity adjustment; its hardening battery is in Supplementary Tables S8 (strict pairing), S7 (denominator sensitivity), and S9 (hydropower exclusion and leave-one-out inference).

| Subset                                        | n treat | ITT    |       |        |        |        | Perm.<br>p |
|-----------------------------------------------|---------|--------|-------|--------|--------|--------|------------|
|                                               |         | ITT    | p     | Down   | Up     | D      |            |
| carryover reservoirs (capacity $\geq 0.5$ yr) | 7       | -0.323 | 0.247 | -0.459 | -0.227 | -0.232 | 0.014      |
| seasonal reservoirs (capacity $< 0.5$ yr)     | 10      | -0.078 | 0.601 | 0.001  | -0.178 | 0.178  | 0.258      |

**Supplementary Table S51 | Minimum detectable effect for the cropland pre-trend.** The flat pre-2010 differential cropland trend (Fig. 4b) is held to the same equivalence standard as the headline nulls: the minimum detectable pre-window trend slope is built from the permutation-null SD (80% power, two-sided alpha 0.05) and cumulated over the 2000–2009 pre-window as a share of the weighted control cropland level. The design would detect a pre-existing divergence accumulating to about half of the control cropland level by 2009; the observed differential trend is about a seventh of the minimum detectable slope.

| Quantity                                                                 | Value      | Detail                                                   |
|--------------------------------------------------------------------------|------------|----------------------------------------------------------|
| pre-2010 differential cropland trend (treat:year, basin fraction per yr) | - 0.000195 | perm p = 0.72 (1000 perms); window 2000-2009             |
| pre-trend MDE (80% power, alpha = 0.05, basin fraction per yr)           | 0.001400   | permutation-null SD 0.0005                               |
| cumulative detectable pre-2010 divergence (% of control cropland level)  | 51.300000  | MDE x 9 yr against control weighted mean fraction 0.0245 |



**Supplementary Table S52 | Storage-band trends and megadrought level shift.** Top: per-year trend slope of the cross-reservoir peak, trough, and peak-trough amplitude (percent of capacity per year; reservoir fixed effects, cluster-robust). Peak and trough both decline while the amplitude trend is not distinguishable from zero, consistent with a supply-side level decline rather than a loss of refill capacity. Bottom: because the storage dynamics are non-linear over a window that begins just before the megadrought, the linear slope is reported as a summary rate only, and the decline is additionally characterized as a within-reservoir level shift at the 2010 onset (pre versus post means with reservoir fixed effects): peak and trough step down significantly while the amplitude shift is indistinguishable from zero.

| Component | Trend (%/yr) | 95% CI          | p     |
|-----------|--------------|-----------------|-------|
| peak      | -1.281       | [-2.21, -0.349] | 0.013 |
| trough    | -1.220       | [-1.72, -0.72]  | 0.000 |
| amplitude | -0.061       | [-1.13, 1.01]   | 0.912 |

| Component | Pre-2010 mean | Post-2010 mean | Shift  | 95% CI            | p       |
|-----------|---------------|----------------|--------|-------------------|---------|
| peak      | 0.867         | 0.665          | -0.210 | [-0.364, -0.0555] | 0.01400 |
| trough    | 0.395         | 0.248          | -0.150 | [-0.227, -0.0734] | 0.00081 |
| amplitude | 0.473         | 0.417          | -0.059 | [-0.217, 0.0987]  | 0.47000 |

**Supplementary Table S53 | Storage-band step-versus-trend decomposition.** The storage decline nested at the 2010 megadrought onset: a level step at 2010 plus a post-2010 drift, for the cross-reservoir peak and trough (reservoir fixed effects, cluster-robust). The peak steps down at the onset ( $-0.19$  of capacity,  $p = 0.04$ ) and shows no further drift ( $p = 0.49$ ); the trough step is not significant ( $p = 0.12$ ) but the trough keeps falling through the megadrought ( $-0.010 \text{ yr}^{-1}$ ,  $p = 0.007$ ). The full-window linear trend of Supplementary Table S52 therefore mixes a level shift into an apparent drift: the peak rests at a lower equilibrium while the carryover floor continues to erode.

| Component | Quantity                          | Estimate | SE    | p      |
|-----------|-----------------------------------|----------|-------|--------|
| peak      | within-megadrought trend (per yr) | -0.002   | 0.003 | 0.4900 |
| trough    | within-megadrought trend (per yr) | -0.010   | 0.003 | 0.0065 |
| peak      | step change at 2010               | -0.192   | 0.088 | 0.0400 |
| peak      | post-2010 drift (per yr)          | -0.002   | 0.003 | 0.4900 |
| trough    | step change at 2010               | -0.074   | 0.046 | 0.1200 |
| trough    | post-2010 drift (per yr)          | -0.010   | 0.003 | 0.0065 |

**Supplementary Table S54 | Carryover capacity of the treated reservoirs.** Whether the 12-month null tests a function the dams were designed to perform is answerable from the project's own data. The median treated reservoir stores about 0.3 years of mean annual downstream flow, seasonal-scale by design, but seven of the 17 reservoirs with a gauged downstream unit hold at least half a year and four hold more than a full year of flow; and seasonal operation does not

empty the fleet, whose pre-2010 annual trough averaged 39% of capacity, so storage persisted across years. Interannual carryover was therefore physically available to a substantial subset, and the multi-year null is not a deterministic consequence of purely seasonal design; the irrigation-only restriction (Supplementary Table S1) additionally shows the result is not diluted by reservoir purpose. Downstream flow is regulated, so the ratios are approximate.

| Quantity                                                        | Value | Detail                                                                                                                          |
|-----------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------|
| reservoir capacity / mean annual downstream flow (median ratio) | 0.29  | IQR [0.15, 0.73]; n = 17 reservoirs with a gauged downstream unit; downstream flow is regulated, so ratios are approximate      |
| reservoirs with carryover ratio $\geq 0.5$                      | 7.00  | of 17; 4 have ratio $\geq 1$ (storage exceeding a full year of flow)                                                            |
| pre-2010 mean annual storage trough (fraction of capacity)      | 0.39  | seasonal drawdown does not empty the reservoirs; storage persists across years, so carryover buffering was physically available |